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Inventor(s)	Park; Kyungwook et al.

Refrigerator

Abstract

A refrigerator includes an inner case including (i) a refrigerating case that defines a refrigerator compartment and (ii) a freezing case that defines a freezer compartment, an outer case disposed outside the inner case, a connection duct disposed between the inner case and the outer case, and a refrigerator compartment cold air supply duct that is detachably coupled to a rear inner surface of the refrigerating case and in fluid communication with the connection duct. The connection duct and the refrigerator compartment cold air supply duct are in communication on a rear surface of the refrigerating case with the refrigerating case interposed therebetween, thereby reducing the area occupied by the refrigerator compartment cold air supply duct disposed in the rear inner surface of the refrigerator compartment.

Inventors: **Park; Kyungwook (Seoul, KR), Kim; Kihwang (Seoul, KR), Park; Jongkwon (Seoul, KR)**

Applicant: **LG ELECTRONICS INC.** (Seoul, KR)

Family ID: **1000008781301**

Assignee: **LG Electronics Inc. (Seoul, KR)**

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Primary Examiner: Zerphey; Christopher R

Attorney, Agent or Firm: Fish & Richardson P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
(1) This application claims priority to and the benefit of Korean Patent Application No. 10-2022-0013687, filed on Jan. 28, 2022, the disclosure of which is incorporated herein by reference in its entirety.
BACKGROUND
Technical Field
(2) The present disclosure relates to a refrigerator, more particularly, a refrigerator that may increase an inner volume of a refrigerator compartment.
Background of the Disclosure
(3) A refrigerator is a home appliance configured to supply cold air generated by refrigerant circulation to a storage chamber (e.g., a refrigerator compartment or a freezer compartment) to keep various kinds of storage targets fresh for a long time in the storage chamber.
(4) A refrigerator compartment refrigerates the storing targets and the freezer compartment freezes the storing target. Due to this structure, the amount of supplied cold air needs to be adjusted differently so that the refrigerator compartment and the freezer compartment may maintain different temperatures.
(5) A refrigerant circulating in the order of a compressor, a condenser, an evaporator and a compressor flows into an evaporator, and the liquid refrigerant is vaporized into gaseous refrigerant. During this process, the cold air supplied to the refrigerator compartment and the freezer compartment may be generated by taking heat from the inside of the refrigerator.
(6) Accordingly, separate evaporators for generating cold air may be provided in the refrigerator compartment and the freezer compartment, respectively, so that the cold air generated by the independent evaporators may be supplied to the compartments, respectively.
(7) When the separate evaporators are provided in the refrigerator compartment and the freezer compartment, respectively, independent cold air supply systems, each of which includes an evaporator for generating cold air, a grill fan assembly for blowing the generated cold air to the refrigerator compartment or the freezer compartment, and a cold air supply duct having a cold air path to supply the cold air, should be also be provided in the refrigerator compartment and the freezer compartment, respectively.
(8) When the separate cold air supply systems mentioned above are provided in the refrigerator compartment and the freezer compartment, respectively, the cold air supplied to the refrigerator and freezer compartments, which require different temperatures and amounts of cold air, may be independently controlled.
(9) However, when the separate cold air systems mentioned above are provided in the refrigerator compartment and the freezer compartment, respectively, there is a problem in that inner volumes of the refrigerator compartment and the freezer compartment are reduced as much as the areas occupied by the cold air supply systems.
(10) For example, the refrigerator compartment has a shape protruding inward so as to sufficiently secure an area in which the separate cold air supply system is disposed at a rear side of the

refrigerator compartment.
(11) As described above, as the area of the region protruding toward the inside of the refrigerator compartment increases more and more, the inner volume of the refrigerator compartment is inevitably reduced by the increased amount.
(12) In addition, the cold air supply duct for supplying the cold air generated by the evaporator to the refrigerator compartment may be formed to have a predetermined thickness to cover the rear surface of the refrigerator compartment.
(13) The cold air supply duct may be disposed on an inner rear surface of the refrigerator compartment. The inner volume of the refrigerator compartment is inevitably reduced as much as the area occupied by the cold air supply duct having the predetermined thickness mounted on the inner rear surface of the refrigerator compartment.
(14) In particular, the cold air supply duct having the cold air path through which cold air passes may include an insulating material having a sufficient thickness to prevent heat exchange with the refrigerator compartment in a relatively humid environment.
(15) To increase the inner volume of the refrigerator compartment, it is necessary to reduce the total area and thickness occupied by the cold air supply duct disposed in the rear surface of the refrigerator compartment.
(16) Meanwhile, the cold air generated by one evaporator may be supplied to the refrigerator compartment and the freezer compartment, without the separate evaporators provided in the refrigerator compartment and the freezer compartment, respectively.
(17) Accordingly, the evaporator for generating cold air and the grill fan assembly for blowing the generated cold air to the refrigerator compartment and the freezer compartment may be disposed in the freezer compartment.
(18) The cold air supply duct having the cold air path to supply the cold air generated in the freezer compartment to the refrigerator compartment may be disposed inside the refrigerator compartment.
(19) In this instance, since the cold air needs to be supplied to the refrigerator and freezer compartments that require different temperatures and amounts of cold air through one evaporator and one grill fan assembly, a flow path opening/closing damper may be further provided in the refrigerator compartment to selectively block the cold air supplied to the refrigerator compartment.
(20) For example, the amount of the cold air supplied to the refrigerator compartment may be adjusted by selectively opening and closing the flow path opening/closing damper disposed in the cold air supply duct.
(21) Accordingly, the refrigerator compartment may have a shape protruding inwardly to sufficiently secure an area in which the flow path opening/closing module is disposed on the rear surface of the refrigerator compartment.
(22) In particular, when the refrigerator compartment is partitioned off into a plurality of spaces that require different temperatures, a plurality of flow path opening/closing modules need to be provided to adjust the mounts of the cold air supplied to the divided spaces, respectively. Accordingly, the refrigerator compartment has a shape further protruding inwardly.
(23) When the flow path opening/closing module is disposed in the refrigerator compartment, the shape protruding inward with respect to the refrigerator compartment may occupy a large area.
(24) As described above, as the area of the region protruding to the inside of the refrigerator compartment increases more and more, the inner volume of the refrigerator compartment is inevitably reduced by that increased amount.
(25) The cold air flow path to the refrigerator compartment from the freezer compartment is blocked by closing the flow path opening/closing damper disposed in the refrigerator compartment, the cold air inside the freezer compartment may be replaced with the cold air inside the relatively humid refrigerator compartment based on the flow path opening/closing damper.
(26) In this instance, since the flow path opening/closing damper is positioned in the refrigerator compartment, the cold air inside the freezer compartment could rise to a rear surface of the

refrigerator compartment having a relatively humid environment, which might cause a problem of dew condensation near the flow path opening/closing damper.
(27) To solve the problem of dew condensation, an insulating material may be reinforced to prevent heat exchange between the cold air of the freezer compartment, which rises up to the area of the flow path opening/closing damper, and the refrigerator compartment.
(28) However, in this instance, there could be another problem in that the inner volume of the refrigerator compartment is additionally reduced by the added thickness of the reinforced insulating material.
(29) Recently, an ice-making chamber may be provided in a refrigerator compartment door so that a user can take out ice without opening the freezer compartment to quickly and easily take out ice without loss of cold air from the freezer compartment.
(30) In general, the cold air supplied to the ice-making chamber of the refrigerator compartment door may be supplied by an evaporator disposed in a rear surface of the freezer compartment to generate cold air.
(31) In order to supply the cold air generated by the evaporator disposed in the rear surface of the freezer compartment to the ice-making chamber disposed in a front surface of the refrigerator, a side cold air supply duct passing through one lateral surface of the refrigerator may be connected to the ice-making chamber to supply cold air.
(32) Since the cold air supplied to the ice-making chamber has a very low temperature, the area near the lateral surface of the refrigerator, through which the side cold air supply duct passes, may have a large temperature difference from another area near the other lateral surface through which the side cold air supply duct does not passes.
(33) As described above, when the temperature difference between the area near one lateral surface of the refrigerator and the area near the other lateral surface becomes large, the overall cold air balance of the refrigerator might deteriorate. Accordingly, there might be difficulties in balancing the overall cold air of the refrigerator and providing an insulation function.
(34) In case of a conventional refrigerator, a suction fan and a filter may be separately installed in a rear surface of a refrigerator compartment to deodorize the refrigerator compartment. The suction fan may be operated to suck and allow cold air inside the refrigerator to pass through the filter, thereby obtaining an effect of deodorizing the refrigerator compartment.
(35) In this instance, to obtain the effect of deodorizing the refrigerator compartment, there is inconvenience of installing a separate suction fan in the refrigerator compartment. If the suction malfunctions, there could be a problem in that the suction fan needs to be repaired separately.
(36) Since a separate space in which the function fan is inserted has to be sufficiently secured in the rear surface of the refrigerator, the area in which the suction fan is installed could greatly protrude inward inside the refrigerator compartment only to reduce the inner volume of the refrigerator.
(37) In addition to the suction fan and filter for deodorizing the refrigerator compartment, a separate suction fan and a separate filter for deodorizing the freezer compartment must be provided in the freezer compartment. There are also a problem of inconvenience of separate management and installation and another problem of inner volume reduction of the freezer compartment.

SUMMARY

(38) One objective of the present disclosure is to provide a refrigerator that may increase an inner volume of a refrigerator compartment by reducing an area occupied by a refrigerator compartment cold air supply duct disposed in an inner rear surface of the refrigerator compartment.
(39) Another objective of the present disclosure is to provide a refrigerator that may increase an inner volume of a refrigerator compartment by reducing an area of a protrusion protruding inward inside the refrigerator compartment by reducing the number of components related to a cold air supply system disposed on an outer rear surface of the refrigerator compartment.
(40) A further objective of the present disclosure is to provide a refrigerator including a new cold air supply system configured to supply cold air generated by one evaporator disposed in a freezer

compartment to a refrigerator compartment having a plurality of divided spaces, the freezer compartment and even an ice-making chamber.
(41) A still further objective of the present disclosure is to provide a refrigerator that may enhance workability by quickly and easily combining a refrigerator compartment cold air supply duct formed in a module assembly shape to an inner surface of the refrigerator compartment.
(42) A still further objective of the present disclosure is to provide a refrigerator that may reduce dew condensation near a flow path opening/closing damper configured to selectively block the cold air generated by an evaporator disposed in a freezer compartment from being supplied to a refrigerator compartment.
(43) A still further objective of the present disclosure is to provide a refrigerator that may balance overall cold air therein, when an ice-making chamber is provided in a front surface of the refrigerator.
(44) A still further objective of the present disclosure is to provide a refrigerator that may increase an inner volume of a refrigerator compartment by reducing an area protruding toward the inside of the refrigerator compartment, while there is no need to install a separate suction fan.
(45) Aspects according to the present disclosure are not limited to the above ones, and other aspects and advantages that are not mentioned above can be clearly understood from the following description and can be more clearly understood from the embodiments set forth herein.
(46) A refrigerator according to an embodiment of the present disclosure is characterized in that a cold air flow path formed by communication between a connection duct and a refrigerator compartment cold air supply duct is extended in upward and downward direction.
(47) The refrigerator may include an inner case comprising a refrigerating case defining a refrigerator compartment and a freezing case defining a freezer compartment; an outer case disposed outside the inner case; an insulating member and a connection duct that are disposed in a space formed between the inner case and the outer case; and a refrigerator compartment cold air supply duct detachably coupled to a rear inner surface of the refrigerating case to be in communication with the connection duct. A cold air flow path formed by communication between the connection duct and the refrigerator compartment cold air supply duct may be extended in upward and downward direction.
(48) The connection duct may be directly in contact with the insulating material.
(49) The cold air flow path may have a linear shape.
(50) A front surface of the refrigerator compartment cold air supply duct and a front surface of the connection duct may include respective predetermined areas parallel to each other.
(51) A refrigerator compartment cold air supply communication hole may be disposed on a rear surface of the inner case and opened upward.
(52) A rear projected portion may be projected from a rear surface of the refrigerating case toward an inside of the refrigerating case, and the cold air supply communication hole may be provided on an upper surface of the rear projected portion.
(53) At least predetermined area of the connection duct may be inserted in the rear projected portion from an outer surface of the refrigerating case.
(54) The connection duct may be secured to the refrigerating case to support a predetermined area of a rear surface of the rear projected portion.
(55) The connection duct may include a rear extended portion vertically extended along a rear end of an upper surface of the connection duct, and the rear extended portion may be secured to the refrigerating case to support a rear surface of the refrigerating case.
(56) An upper surface of the connection duct and a lower surface of the refrigerator compartment cold air supply duct may include inclined surfaces, respectively, and the upper surface of the connection duct may face the lower surface of the refrigerator compartment cold air supply duct.
(57) The inclined surface may descend toward a front surface of the refrigerator compartment, and the upper surface of the connection duct and the lower surface of the refrigerator compartment cold

air supply duct may be coupled to each other by sliding.
(58) A duct inserting groove may be provided on an upper surface of the refrigerating case and an upper area of the refrigerator compartment cold air supply duct may be partially inserted in the duct inserting groove to be secured.
(59) The duct inserting groove may be disposed along an area in which an upper surface and a rear surface of the refrigerating case meet each other.
(60) The refrigerator compartment cold air supply duct may include a duct insulating portion in which a cold air flow path is formed; and a duct sheet coupled to the duct insulating portion in a rear area of the duct insulating portion, and the duct sheet has a thickness that is smaller than a thickness of the duct insulating portion.
(61) One end of the connection duct may be connected to the freezer compartment and the other end of the connection duct may be connected to the refrigerator compartment to pass through a center of the refrigerator compartment in a left-right direction. the width of the cold air flow path may increase from one end to the other end of the connection duct.
(62) The refrigerator may further include a door configured to open and close the refrigerator compartment; an ice-making chamber provided in the door; and an ice-making chamber cold air supply duct configured to supply cold air to the ice-making chamber. The ice-making chamber cold air supply duct may be disposed to pass through the other lateral surface facing one lateral surface of the refrigerator compartment adjacent to the connection duct.
(63) The refrigerator compartment cold air supply duct may include a first refrigerator compartment cold air flow path and a second refrigerator compartment cold air flow path that may be configured to branch the cold air guided from the connection duct. The first refrigerator compartment cold air flow path may have a flow path width that is wider than the second refrigerator compartment cold air flow path. The second refrigerator compartment cold air flow path may be disposed closer to the ice-making chamber cold air supply duct than the first refrigerator compartment cold air flow path.
(64) The refrigerator may further comprise a refrigerator compartment cold air returning duct configured to return cold air inside the refrigerator compartment. One end and the other end of the refrigerator compartment cold air returning duct may be connected to the refrigerator compartment and the freezer compartment, respectively. One end and the other end of the refrigerator compartment cold air returning duct may be positioned to pass through a center of the refrigerator compartment and a center of the freezer compartment with respect to a left-right direction.
(65) The refrigerator compartment may include a first storage chamber and a second storage chamber. The refrigerator compartment may further include a second storage chamber cold air supply duct disposed on an outer surface of the refrigerating case and configured to supply cold air to the second storage chamber. The connection duct and the second storage chamber cold air supply duct may be disposed in an area of the refrigerator compartment cold air returning duct, and the ice-making chamber cold air supply duct may be disposed in the other area of the refrigerator compartment cold air returning duct with respect to a left-right direction.
(66) The second storage chamber cold air supply duct may be directly in contact with the insulating material.
(67) A first storage chamber flow path opening/closing damper configured to selectively cut off the cold air supplied to the connection duct and a second storage chamber flow path opening/closing damper configured to selectively cut off the cold air supplied to the second storage chamber cold air supply duct may be provided in the freezer compartment.
(68) The refrigerator compartment cold air supply duct may include a refrigerator compartment cold air main outlet hole configured to discharge cold air to the refrigerator compartment; and a refrigerator compartment cold air auxiliary outlet hole. A refrigerator compartment cold air auxiliary outlet guide comprising a refrigerator compartment filter may be provided on a front surface of the refrigerator compartment cold air auxiliary outlet hole, and the cold air supplied

through the refrigerator compartment cold air auxiliary outlet hole may pass through the refrigerator compartment filter.
(69) The refrigerator compartment cold air auxiliary outlet guide may further include an insulating material spaced a preset distance apart from a front surface of the refrigerator compartment filter, and the refrigerator compartment cold air auxiliary outlet guide may be configured to guide the cold air after passing through the refrigerator compartment filter to be discharged downward inside the refrigerator compartment through a space between the insulating material and the refrigerator compartment filter.
(70) A returning duct filter may be disposed inside the refrigerator compartment cold air returning duct, and the cold air returning by the refrigerator compartment cold air returning duct may pass through the returning duct filter.
(71) The connection duct and the refrigerator compartment cold air supply duct are in communication on a rear surface of the refrigerating case with the refrigerating case interposed therebetween, thereby reducing the area occupied by the refrigerator compartment cold air supply duct disposed in the rear inner surface of the refrigerator compartment.
(72) The refrigerator according to the present disclosure may enhance capacity competitiveness by increasing the inner volume of the refrigerator compartment.
(73) Furthermore, in the refrigerator according to the present disclosure, the flow path opening/closing module configured to selectively cut off the cold air supplied to the refrigerator compartment may be disposed in the freezer compartment, thereby reducing components related to the cold air supply system disposed on the outer rear surface of the refrigerator compartment.
(74) Accordingly, the refrigerator may increase the inner volume of the refrigerator compartment by reducing the area of the projected part projected toward the inside of the refrigerator compartment and then enhance the capacity competitiveness.
(75) Still further, the refrigerator according to the present disclosure may supply cold air to all of the refrigerator compartment, the freezer compartment and the ice-making chamber by using one evaporator and one grill fan assembly, and may adjust the cold air supplied to the refrigerator compartment divided into the first storage chamber and the second storage chamber by using the first storage chamber flow path opening/closing damper and the second storage chamber opening/closing damper provided in the grill fan assembly.
(76) Accordingly, the refrigerator may provide a new cold air supply system capable of adjusting the cold air supply to the refrigerator compartment divided into the two storage chambers, the freezer compartment and the ice-making chamber by using one evaporator and the plurality of flow path opening/closing dampers provided in the freezer compartment.
(77) Still further, in the refrigerator according to the present disclosure, the upper surface of the connection duct and the lower surface of the refrigerator compartment cold air supply duct, which have the inclined surfaces, respectively, may be coupled to each other by sliding, while pushing the refrigerator compartment cold air supply duct upward after inserting some area of the upper region of the refrigerator compartment cold air supply duct.
(78) Accordingly, the refrigerator compartment cold air supply duct may be easily coupled inside the refrigerator compartment as a module assembly unit. The upper region of the refrigerator compartment cold air supply duct may be inserted in the duct inserting groove even without separate fastening means to be strongly secured by surface-to-surface contact, thereby facilitating assembling process simplification and improving workability.
(79) Still further, in the refrigerator according to the present disclosure, the flow path opening/closing damper may be disposed in the freezer compartment, not the refrigerator compartment with a relatively humid environment. Accordingly, when the flow path opening/closing damper is closed, cold air inside the freezer compartment may not rise to the refrigerator compartment but stay in the freezer compartment, thereby reducing dew condensation near the flow path opening/closing damper.

(80) Still further, the refrigerator according to the present disclosure may have the cold air supply system structure in which the connection duct and the second storage chamber cold air supply duct, which supply cold air to the refrigerator compartment, is disposed adjacent to one lateral surface of the refrigerator and the ice-making chamber cold air supply duct for supplying cold air to the ice-making chamber is disposed adjacent to the other lateral surface of the refrigerator.
(81) Accordingly, the refrigerator may provide a new cold air supply system capable of balancing the entire cold air inside the refrigerator.
(82) Still further, the refrigerator according to the present disclosure may perform the deodorizing function for the refrigerator compartment by using the refrigerator compartment cold air auxiliary flow path branched from the refrigerator cold air flow path for supplying cold air to the refrigerator compartment, thereby having no need of a separate suction fan provided in the refrigerator compartment to deodorize the refrigerator compartment.
(83) Accordingly, the area projected to the inside of the refrigerator compartment to install the suction fan may be reduced and then the inner volume of the refrigerator compartment may be increased. The deodorizing function for the refrigerator compartment may be performed without interfering with the circulation structure of the cold air inside the refrigerator compartment.
(84) Specific effects are described along with the above-described effects in the section of Detailed Description.

Description

DESCRIPTION OF REFERENCE NUMERALS
(1) FIGS. **1**A and **1**B are front perspective views showing a state where a door of a refrigerator including an ice-making chamber is closed and a state where a door of a refrigerator including no ice-making chamber is closed;
(2) FIG. **2** is a front perspective view showing a state where a door of a refrigerator is open;
(3) FIGS. **3** and **4** are front and rear perspective views showing a state where an inner case, various ducts and a grill fan assembly are coupled to each other;
(4) FIG. **5** is a rear perspective view showing a state where various ducts and a grill fan assembly are coupled to each other;
(5) FIG. **6** is a rear perspective view of an inner case;
(6) FIG. **7** is a front view of a refrigerator having internal components removed;
(7) FIG. **8** is a front view of a refrigerator including a grill fan assembly and a refrigerator compartment cold air duct;
(8) FIG. **9** is a front view of a refrigerator including a grill fan assembly and a refrigerator compartment cold air supply duct, which shows a perspective-viewed inside;
(9) FIG. **10** is a rear view of a refrigerating case to which various ducts are coupled;
(10) FIG. **11** is a view of a second storage chamber cold air outlet cover coupled to a second storage chamber cold air supply duct;
(11) FIGS. **12** and **13** are a side sectional perspective view and a side sectional view that show a connection structure between a refrigerator compartment cold air supply duct and a connection duct;
(12) FIG. **14** is a front perspective view showing a cold air circulation structure of a refrigerator compartment;
(13) FIG. **15** is a front perspective view showing a cold air circulation structure of a second storage chamber;
(14) FIG. **16** is a front perspective view showing a cold air circulation structure of a freezer compartment;
(15) FIG. **17** is a front perspective view showing a cold air circulation and returning structure of an

ice-making chamber;
(16) FIG. **18** is a rear perspective view showing a refrigerator compartment cold air supply duct coupled to a refrigerating case;
(17) FIG. **19**A is a front perspective view of a refrigerator compartment cold air supply duct, FIG. **19**B is a rear perspective view of a refrigerator compartment cold air supply duct, FIG. **19**C is a rear exploded perspective view showing some parts of a refrigerator compartment cold air supply duct, FIG. **19**D is a front exploded perspective view of a refrigerator compartment cold air supply duct, and FIG. **19**E is a rear exploded perspective view of a refrigerator compartment cold air supply duct.
(18) FIG. **20**A is a rear exploded perspective view showing some parts of a refrigerator compartment cold air supply duct including a refrigerator compartment filter, FIG. **20**B is a front view showing a refrigerator compartment cold air auxiliary outlet hole, and FIG. **20**C is a side sectional view of a refrigerator compartment cold air auxiliary outlet hole to which a refrigerator compartment filter is coupled;
(19) FIG. **21**A is a perspective view showing a state where a refrigerator compartment cold air supply duct, a connection duct and a second storage chamber cold air supply duct are coupled to each other;
(20) FIG. **21**B is a perspective view showing a state where a refrigerator compartment cold air supply duct, a connection duct and a second storage chamber cold air supply duct are decoupled from each other;
(21) FIGS. **22**A and **22**B are a front perspective view and a rear perspective view that show a connection duct and a second storage chamber cold air supply duct are coupled to each other;
(22) FIG. **22**C is an exploded perspective view showing a connection duct and a second storage chamber cold air supply duct;
(23) FIGS. **22**D and **22**E are a rear view and a front view showing a state where a connection duct and a second storage chamber cold air supply duct are coupled to each other;
(24) FIGS. **22**F, **22**G, **22**H and **22**I are a left sectional view, a right sectional view, a front view and a bottom view showing a state where a connection duct and a second storage chamber cold air supply duct are coupled to each other;
(25) FIGS. **23**A and **23**B are a front perspective view and a rear perspective view showing a refrigerator compartment cold air returning duct;
(26) FIG. **23**C is an exploded perspective view of a refrigerator compartment cold air returning duct, and FIG. **23**D is a right sectional view of a refrigerator compartment cold air returning duct;
(27) FIG. **24**A is a perspective view showing an upper area of a grill fan assembly including two flow path opening/closing dampers, FIG. **24**B is a front view of a grill fan assembly, FIG. **24**C is a front view of a grill fan assembly having a grill fan removed, FIG. **24**D is an exploded perspective view of a grill fan assembly, and FIG. **24**E is a rear view a state where an evaporator is disposed on a rear surface of a grill fan assembly;
(28) FIG. **25** is a perspective view of a grill fan assembly including one flow path opening/closing damper;
(29) FIGS. **26**A and **26**B are a rear perspective view and an exploded perspective view of a grill assembly including one flow path opening/closing damper and an ice-making chamber; and
(30) FIG. **27**A and FIG. **27**B are a rear perspective view and an exploded perspective view of a grill fan assembly including one flow path opening/closing damper and no ice-making chamber.
DESCRIPTION OF SPECIFIC EMBODIMENTS
(31) The above-described aspects, features and advantages are specifically described hereunder with reference to the accompanying drawings such that one having ordinary skill in the art to which the present disclosure pertains can easily implement the technical spirit of the disclosure. In the disclosure, detailed descriptions of known technologies in relation to the disclosure are omitted if they are deemed to make the gist of the disclosure unnecessarily vague. Below, preferred

embodiments according to the disclosure are specifically described with reference to the accompanying drawings. In the drawings, identical reference numerals can denote identical or similar components.
(32) The terms "first", "second" and the like are used herein only to distinguish one component from another component. Thus, the components should not be limited by the terms. Certainly, a first component can be a second component unless stated to the contrary.
(33) Throughout the disclosure, each component can be provided as a single one or a plurality of ones, unless explicitly stated to the contrary.
(34) Hereinafter, expressions of 'a component is provided or disposed in an upper or lower portion' may mean that the component is provided or disposed in contact with an upper surface or a lower surface. The present disclosure is not intended to limit that other elements are provided between the components and on the component or beneath the component.
(35) It will be understood that when an element is referred to as being "connected with" another element, the element can be directly connected with the other element or intervening elements may also be present. In contrast, when an element is referred to as being "directly connected with" another element, there are no intervening elements present.
(36) A singular representation may include a plural representation unless it represents a definitely different meaning from the context. Terms such as "include" or "has" are used herein and should be understood that they are intended to indicate an existence of several components, functions or steps, disclosed in the specification, and it is also understood that greater or fewer components, functions, or steps may likewise be utilized.
(37) Throughout the disclosure, the terms "A and/or B" as used herein can denote A, B or A and B, and the terms "C to D" can denote C or greater and D or less, unless stated to the contrary.
(38) Hereinafter, a refrigerator according to several embodiments will be described.
(39) [Overall Structure of Refrigerator]
(40) FIGS. **1**A and **1**B are front perspective views showing a state where a door of a refrigerator including an ice-making chamber is closed and a state where a door of a refrigerator including no ice-making chamber is closed. FIG. **2** is a front perspective view showing a state where a door of a refrigerator is open.
(41) An exterior design of the refrigerator **1** may be defined by a cabinet **1** defining a storage space and a door configured to open and close an open front of the cabinet **2**.
(42) The cabinet **2** may include an outer case **10** forming an outer surface of the refrigerator **1** and an inner case **40** forming an inner surface of the outer case **10**.
(43) The outer case **10** and the inner case **40** may be spaced a preset distance apart from each other and an insulating material is foamed in the space between them to fill the empty space with the insulating material.
(44) A storage space inside the cabinet **2** may be divided into a plurality of spaces, which are a refrigerator compartment **51** and a freezer compartment **52**.
(45) As one embodiment of the present disclosure, the freezer compartment **52** may be mounted in a lower space of the cabinet **2** and the refrigerator compartment **51** may be mounted in an upper space.
(46) A door may be coupled to a front surface of the cabinet **2** to open and close the refrigerator **1**.
(47) An upper door **20** may be coupled to a front surface corresponding to the refrigerator compartment **51** and a lower door **30** may be coupled to a front surface corresponding to the freezer compartment **52**.
(48) For example, the upper door **20** may be a rotation type configured of a first upper door **20** and a second upper door **20***b* that are rotatable on shafts on both sides of the cabinet **2**, respectively.
(49) The lower door **30** may be a drawer type configured to slide inward or outward along a rail.
(50) Referring to FIG. **1**A, a dispenser **21** may be disposed in the first upper door **20***a* and configured to discharge water or ice even when the door is not opened.

(51) Referring to FIG. **2**, an ice-making chamber **22** may be disposed in the first upper door **20***a* in which the dispenser **21** is provided, and may be configured to make ice.
(52) On an inner surface of the inner case **40** connected to the first upper door **20***a* may be formed an ice-making chamber cold air supply outlet hole **600***b* for supply cold air to the ice-making chamber **22** and an ice-making cold air returning inlet hole **700***a* for returning cold air from the ice-making chamber **22**.
(53) The ice-making chamber cold air supply outlet hole **600***b* and the ice-making cold air returning inlet hole **700***a* may be in communication with one surface of the ice-making chamber **22**, in a state where the first upper door **20***a* is closed.
(54) The refrigerator compartment **51** may be divided into a first storage chamber **51***a* and a second storage chamber **51***b*.
(55) The second storage chamber **51***b* may be a pantry room that may control the temperature to accommodate a specific storage target such as vegetables or meat.
(56) The first storage chamber **51***a* may refer the other space of the refrigerator compartment **51**, except the second storage chamber **51***b*, and may be a main storage space.
(57) For example, the second storage chamber **51***b* may be disposed below the first storage chamber **51***a*, and may be partitioned off as a separate space from the first storage chamber **51***a* by a partitioning member.
(58) A storage drawer **3** may be provided in the second storage chamber **51***b* and configured to slide outward and inward along a rail.
(59) In addition, a storage drawer **3** or a shelf **4** may be provided in the first storage chamber **51***a* to easily keep or preserve fresh storing targets.
(60) Separate temperature sensors may be provided in the first storage chamber **51***a* and the second storage chamber **51***b*, respectively, and configured to independently adjust and keep different temperatures.
(61) [Cold Air Supply System and Connection Relation Between Components]
(62) Hereinafter, referring to FIGS. **3** to **10**, a new cold air supply system formed by coupling the inner case, various ducts and a grill fan assembly to each other and the connection relation between them will be described.
(63) FIGS. **3** and **4** are front and rear perspective views showing a state where an inner case, various ducts and a grill fan assembly are coupled to each other. FIG. **5** is a rear perspective view showing a state where various ducts and a grill fan assembly are coupled to each other. FIG. **6** is a rear perspective view of an inner case.
(64) FIG. **7** is a front view of a refrigerator having internal components removed. FIG. **8** is a front view of a refrigerator including a grill fan assembly and a refrigerator compartment cold air duct. FIG. **9** is a front view of a refrigerator including a grill fan assembly and a refrigerator compartment cold air supply duct, which shows a perspective-viewed inside. FIG. **10** is a rear view of a refrigerating case to which various ducts are coupled.
(65) The inner case **40** may include a refrigerating case **41** disposed in an upper area and constituting the refrigerator compartment **51**, and a freezing case **42** disposed in a lower area and constituting the freezer compartment **52**.
(66) The cold air generated by one evaporator **101** may be supplied both of the refrigerator compartment **51** and the freezer compartment **52**.
(67) When the ice-making chamber **22** is additionally provided in the upper door **20** of the refrigerator **1**, the cold air generated by one evaporator **101** may be supplied to all of the refrigerator compartment **51**, the freezer compartment **52** and the ice-making chamber **22**.
(68) The evaporator **101** for generating cold air may be disposed inside the freezer compartment **52**, specifically, on a rear surface **42***a* of the freezing case **42**.
(69) The evaporator **101** may be disposed in an upper area of a mechanical chamber **53**.
(70) The mechanical chamber **53** may be provided in a rear lower area of the freezing case **42**, and

configured to provide a space in which a compressor **54** and a condenser are installed.
(71) The lower rear space inside the freezer compartment **52** may have a relatively narrower freezing space than an upper rear space inside the freezer compartment **52** by the space occupied by the mechanical chamber **53**.
(72) In other words, an upper surface **42***b* of the freezing case may have a larger area than a lower surface **42***c* of the freezing case.
(73) Accordingly, an upper region of the freezer compartment **52** may more protrude rearward than a lower region of the freezer compartment **52**, so that the evaporator **101** may be disposed in an upper rear space inside the freezer compartment **52**.
(74) A grill fan assembly **100** configured to blow the cold air generated by the evaporator **101** to the refrigerator compartment **51** and the freezer compartment **52** may be disposed on a front surface of the evaporator **101**.
(75) When the ice-making chamber **22** is provided in the upper door **20** of the refrigerator **1**, the cold air generated by one evaporator **101** may be blown from one grill fan assembly **100** to all of the refrigerator compartment **51**, the refrigerator compartment **51** and the ice-making chamber **22**.
(76) Accordingly, the refrigerator **1** according to the present disclosure may supply the cold air generated by one evaporator **101** not only to the freezer compartment **52** but also to the refrigerator compartment **51**.
(77) Since there is no need of securing the space in the refrigerator compartment **51** to accommodate a separate evaporator **101**, the inner volume of the refrigerator compartment **51** may be increased.
(78) To blow cold air to a refrigerator compartment cold air supply duct **300**, a connection duct **200** may be further provided between the grill fan assembly **100** and the refrigerator compartment cold air supply duct **300**.
(79) One end of the connection duct **200** may be connected to the grill fan assembly **100** and the other end of the connection duct **200**, so that the cold air blown from the grill fan assembly **100** may be guided to the refrigerator compartment cold air supply duct **300**.
(80) The refrigerator compartment cold air supply duct **300** may be disposed on an inner surface the refrigerating case **41**, and the connection duct **200** may be disposed on an outer surface of the refrigerating case **41**. The refrigerator compartment cold air supply duct **300** and the connection duct **200** may be in communication at a rear surface **41***a* of the refrigerating case.
(81) An insulating material **11** may be foamed in a space between the inner case **40** and the outer case **10** to fill in the space.
(82) The connection duct **200** may be embedded in the space between the inner case **40** and the outer case **10** by passing through the space foamed and filled with the insulating material **11**.
(83) Accordingly, when the upper door **20** of the refrigerator **1** is opened, the refrigerator compartment cold air supply duct **300** disposed on the inner surface of the refrigerating case **41** may be exposed to the outside but the connection duct **200** disposed on the outer surface of the refrigerating case **41** may not be exposed.
(84) A rear projected portion **43** protruding to the inside of the refrigerating case **41** may be provided on a rear surface **41***a* of the refrigerating case so that at least predetermined area of the connection duct **200** may be inserted from the outside of the refrigerating case **41**.
(85) The rear projected portion **43** may be formed in a shape corresponding to the connection duct **200** to receive the connection duct **200**.
(86) The rear projected portion **43** may have a shape extending from a lower surface **41***c* toward an upper surface **41***b* of the refrigerating case along the rear surface **41***a* of the refrigerating case.
(87) The refrigerator compartment cold air supply duct **300** may be disposed in the refrigerating case **41**. Accordingly, as the area occupied by the refrigerator compartment cold air supply duct **300** increases, the inner volume of the refrigerating case **41** may decrease.
(88) In particular, the refrigerator compartment cold air supply duct **300** has a cold air flow path for

cold air. When the cold air flow path passes through the refrigerator compartment **51** having a relatively humid environment, dew condensation could occur. Accordingly, the refrigerator compartment cold air supply duct **300** may include an insulating material with a predetermined thickness.
(89) To increase the inner volume of the refrigerating case **41**, it is necessary to reduce the area occupied by the refrigerator compartment cold air duct **300** disposed on the inner surface of the refrigerating case **41**.
(90) The refrigerator compartment cold air supply duct **300** according to the present disclosure may extend to the upper surface **41***b* from the lower surface **41***c* of the refrigerating case, not to be disposed on the rear surface **41***a* of the refrigerating case.
(91) The rear projected portion **43** having a shape extending along the rear surface **41***a* of the refrigerating case may be disposed from the lower surface **41***c* of the refrigerating case to a predetermined height toward the upper surface **41***b* of the refrigerating case.
(92) Since the connection duct **200** is disposed on a rear surface of the rear projected portion **43**, the connection duct **200** may be disposed on an outer surface not an inner surface of the refrigerating case **41**.
(93) Accordingly, an additional area protruding toward the inside of the refrigerating case **41** except the rear projected portion **43** may be reduced up to the height of the rear projected portion **43** in which the connection duct **200** is inserted, so that the inner volume of the refrigerating case **41** may be increased by that much.
(94) The rear projected portion **43** may extend up to a height near a center area with respect to a vertical direction of the refrigerating case **41**, but the present disclosure is not limited thereto.
(95) For example, the rear projected portion **43** may extend to from the lower surface **41***c* of the refrigerating case to a height at which the rear projected portion **43** can be covered by the storage drawer **3**.
(96) Alternatively, the rear projected portion **43** may be covered by the second storage chamber **51***b* formed in a lower region of the refrigerating case **41**.
(97) The rear projected portion **43** may be extended from the lower surface **41***c* of the refrigerating case up to the height to be covered by the second storage chamber **51***b* formed in the lower region of the refrigerating case **41**.
(98) The rear projected portion **43** may be projected in the shape corresponding to the connection duct **200**, not a shape protruding evenly from the rear surface **41***a* of the refrigerating case, which is not an aesthetically pleasing shape.
(99) Accordingly, up to the height at which the rear projected portion **43** is formed, the second storage chamber **51***b* and the storage drawer **3** may be disposed in front of the rear projected portion **43** not to expose the rear projected portion **43** even when the upper door **20** is open, thereby increasing the aesthetic effect of the inside of the refrigerator **1**.
(100) In addition, since the connection duct **200** is configured to pass through the space between the inner case **40** and the outer case **10** that is formed and filled with the insulating material **11**, there is no need of an additional insulation material for preventing heat exchange between the refrigerator compartment **51** and the connection duct **200** through cold air passes.
(101) In order to provide a thermal insulation effect to the refrigerator **1**, an insulating material **11** having very low heat conductivity may be foamed between the inner case **40** and the outer case **10** and the space between the inner case **40** and the outer case **10** may be filled.
(102) If the connection duct **200** is provided in the refrigerating case **41**, the connection duct **200** may be insulated by providing an insulating material with a predetermined thickness.
(103) Alternatively, the connection duct **200** may be embedded to pass through the space foamed and filled with the insulating material **11** between the inner case **40** and the outer case **10** as described in the embodiment of the present disclosure.
(104) Accordingly, according to the embodiment, a sufficient insulation effect may be provided by

using the insulating material **11** already foamed in the space between the inner case **40** and the outer case **10**, without adding a separate insulating material for the heat insulation of the connection duct **200**.
(105) As a result, according to the embodiment of the present disclosure, no separate heat insulating material for providing the terminal insulation effect to the connection duct **200** is required so that the thickness of the connection duct **200** may be greatly reduced.
(106) Accordingly, the thickness of the rear projected portion **42** protruding toward the inside of the refrigerating case **41** may be greatly reduced, thereby increasing the inner volume of the refrigerator compartment **51**.
(107) Accordingly, since the connection duct **200** and the refrigerator compartment cold air supply duct **300** are in communication at the rear surface of the refrigerating case, while having the refrigerating case **41** disposed between them, the refrigerator **1** according to the embodiment of the present disclosure may reduce the area occupied by the refrigerator compartment cold air supply duct **300** disposed on the inner rear surface **41***a* of the refrigerating case.
(108) Due to this structure, the refrigerator according to the embodiment may greatly increase the inner volume of the refrigerator compartment **51**.
(109) Hereinafter, additionally referring to FIGS. **12** and **13**, the connection relationship between the refrigerator compartment cold air supply duct **300** and the connection duct **200** will be described in detail.
(110) The connection duct **200** may be fixed to the refrigerating case **41** to support the rear surface of the rear projected portion **43** and the rear surface **41***a* of the refrigerating case.
(111) A rear extended portion **221** may be vertically extended from a rear end of an upper end of the connection duct **200**.
(112) The rear extended portion **221** may be configured to support the rear surface **41***a* of the refrigerating case.
(113) A connection duct securing through hole **223** for securing the connection duct **200** to the rear surface **41***a* of the refrigerating case by using a fastening member such as a bolt may be formed in each side of the rear extended portion **221**.
(114) The connection duct **200** may be primarily fixed to the refrigerating case **41** through the connection duct securing through hole **223** by using the fastening member.
(115) Hence, as the insulating material **11** is additionally foamed and filled in the space between the refrigerating case **41** and the outer case **10**, the connection duct **200** may be secondarily fixed to the refrigerating case **41** to have a stronger fixing force.
(116) The upper surface of the connection duct **200** may be configured to support a rear surface of the rear projected portion **43**, and the rear extended portion **2221** of the connection duct **200** may be configured to support the rear surface **41***a* of the refrigerating case.
(117) Accordingly, the connection duct **200** may provide the refrigerating case **41** with a strong supporting force at the lower surface and the rear surface of the refrigerating case **41** simultaneously.
(118) As described above, the connection duct **200** according to the embodiment may serve not only to transfer the cold air generated by the evaporator **101** to the refrigerator compartment cold air supply duct **300** but also to strongly support the refrigerating case **41** as a support member.
(119) The rear projected portion **43** may include a refrigerator compartment cold air supply communication hole **44** for facilitate communication between the connection duct **200** and the refrigerator compartment cold air supply duct **300**.
(120) The connection duct **200** and the refrigerator compartment cold air supply duct **300** may be in communication through the refrigerator compartment cold air supply communication hole **44**, with the refrigerating case disposed therebetween.
(121) The refrigerator compartment cold air supply communication hole **44** may be provided on the upper surface of the rear projected portion **43**, and the upper surface of the rear projected portion **43**

may have an inclined surface that descends in a direction to the front surface of the refrigerator compartment **51**.

(122) A refrigerator compartment cold air guide outlet hole **200***b* may be provided on the upper surface of the connection duct **200** and the upper surface of the connection duct **200** may have an inclined surface that descends in a direction to the front surface of the refrigerator compartment **51**.

(123) A refrigerator compartment cold air inlet hoe **300***a* may be disposed on the lower surface of the refrigerator compartment cold air supply duct **300**, and the lower surface of the refrigerator compartment cold air supply duct **300** may have an inclined surface that descends in a direction to the front surface of the refrigerator compartment **51**.

(124) A refrigerator compartment cold air supply connecting portion **310** may be extended downward from a lower surface of the refrigerator compartment cold air supply duct **300**.

(125) The refrigerator compartment cold air supply connecting portion **310** may have a shape corresponding to the upper surface of the connection duct **200** to be coupled to the upper surface of the connection duct **200**.

(126) The refrigerator compartment cold air supply connecting portion **310** may have a left-and-right width narrower than the entire left-and-right width of the refrigerator compartment cold air supply duct **300**, and may have a shape protruding downward.

(127) The left-and-right width of the refrigerator compartment cold air supply connecting portion **310** may be substantially equal to that of the upper surface of the connection duct **200** coupled to the refrigerator compartment cold air supply connecting portion.

(128) The refrigerator compartment cold air inlet hole **300***a* formed on the lower surface of the refrigerator compartment cold air supply duct **300** may be formed in the refrigerator compartment cold air supply connecting portion **310** extended downward with respect to the refrigerator compartment cold air supply duct **300**.

(129) An inclined surface that descends in a direction to the front surface of the refrigerator compartment **51**, so that the inclined surface of the rear projected portion **43**, the inclined surface of the upper surface of the connection duct **200** and the inclined surface of the lower surface of the refrigerator compartment cold air supply duct **300** may have substantially the same angle.

(130) The upper surface of the connection duct **200** and the lower surface of the refrigerator compartment cold air supply duct **300** may be coupled to each other so that the inclined surfaces may face each other with the rear projected portion **43** of the refrigerating case **41** interposed therebetween.

(131) Specifically, the inclined surface of the upper surface of the connection duct **200** and the inclined surface of the lower surface of the refrigerator compartment cold air supply duct **300** may be fixedly coupled to each other with the inclined surface of the rear projected portion **43** interposed therebetween.

(132) In this instance, the lower surface of the refrigerator compartment cold air supply duct **300** fixedly coupled to the inclined surface of the upper surface of the connection duct **200** may be the lower surface of the refrigerator compartment cold air supply connecting portion **310**.

(133) A duct bottom cover **305** configured to cover the lower surface of the refrigerator compartment cold air supply connecting portion **310** may be further disposed underneath the refrigerator compartment cold air supply connecting portion **310**.

(134) The lower surface of the refrigerator compartment cold air supply duct **300** fixedly coupled to the inclined surface of the upper surface of the connection duct **200** may be a duct bottom cover **305**.

(135) A refrigerator compartment cold air inlet hole **300***a* may be formed even on the duct bottom cover **305** and configured to receive the cold air guided from the connection duct **200** into the refrigerator compartment cold air supply duct **300**.

(136) The inclined surface of the rear projected portion **43**, the inclined surface of the upper surface of the connection duct **200** and the inclined surface of the lower surface of the refrigerator

compartment cold air supply duct **300** may be fixed coupled to each other, so that the refrigerator compartment cold air guide outlet hole **200***b*, the refrigerator compartment cold air supply communication hole **44** and the refrigerator compartment cold air inlet hole **300***a* may be disposed in communication with each other.
(137) Due to the above-described cold air flow path communication structure, the cold air generated by the evaporator **101** may be supplied to the refrigerator compartment **51** from the freezer compartment **52**.
(138) Meanwhile, a duct inserting groove **49** in which some upper region of the refrigerator compartment cold air supply duct **300** is fixedly inserted may be provided on the upper surface **41***b* of the refrigerating case.
(139) The duct inserting groove **49** may be provided along an area in which the upper surface **41***b* meets the rear surface **41***a* of the refrigerating case.
(140) The duct inserting groove **49** may be formed in a protrusion shape protruding upward, viewed above the upper surface **41***b* of the refrigerating case, or a concave shape recessed upward, viewed below the upper surface **41***b* of the refrigerating case.
(141) The front-rear width of the duct inserting groove **49** may be formed slightly longer than the front-rear width of the upper region of the refrigerator compartment cold air supply duct **300**, so that the upper region of the refrigerator compartment cold air supply duct **300** may be smoothly inserted and fixed in the duct inserting groove **49**.
(142) The upper region of the refrigerator compartment cold air supply duct **300** may be strongly fixed by being inserted in the duct inserting groove **49** to be in surface-to-surface contact.
(143) As described above, the upper surface of the connection duct **200** and the lower surface of the refrigerator compartment cold air supply duct may have the respective inclined surfaces coupled to each other, the inclined surfaces may descend in the direction to the front surface of the refrigerator compartment **51**.
(144) Due to the coupling structure of the inclined surfaces, the upper region of the refrigerator compartment cold air supply duct **300** may be first inserted in the duct inserting groove **49**, and may then couple the refrigerator compartment cold air supply duct **300** while pushing the refrigerator compartment cold air supply duct **300** upward.
(145) In this instance, the lower surface of the refrigerator compartment cold air supply duct **300** coupled to the upper surface of the connection duct **200** may be the lower surface of the refrigerator compartment cold air supply connecting portion **310**.
(146) Accordingly, after some area of the upper region of the refrigerator compartment cold air supply duct **300** is inserted in the duct inserting groove **49**, the upper surface of the connection duct **200** and the lower surface of the refrigerator compartment cold air supply connecting portion **310**, which have the respective surfaces, may be fixedly coupled to each other in a sliding manner while pushing the refrigerator compartment cold air supply duct **200** upward.
(147) As described above, the upper surface of the connection duct **200** and the lower surface of the refrigerator compartment cold air supply duct **300** may be coupled to each other to make the respective inclined surfaces face each other with the rear projected portion **43** of the refrigerating case **41** interposed therebetween.
(148) Accordingly, some area of the upper region of the refrigerator compartment cold air supply duct **300** may be smoothly inserted in the duct inserting groove **49** disposed on the upper surface **41***b* of the refrigerating case to be strongly fixed.
(149) Since the refrigerator compartment cold air supply duct **300** is easily and quickly coupled to the inside of the refrigerator compartment **51** as a module assembly type, workability may be greatly improved.
(150) In addition, the upper region of the refrigerator compartment cold air supply duct **300** may be inserted and strongly fixed in the duct inserting groove **49** to be in surface-to-surface contact, even without separate fastening means, thereby greatly improving process simplification and

workability.
(151) A hook securing portion **211** may be formed on the upper surface of the connection duct **200** slidingly coupled to the lower surface of the refrigerator compartment cold air supply duct **300**, and configured to guide the sliding-coupling.
(152) A hook **314** may be formed on the lower surface of the refrigerator compartment cold air supply connecting portion **310** and configured to be guided and secured to the hook securing portion **211** of the connection duct **200**.
(153) When the duct bottom cover **305** is further disposed on the lower surface of the refrigerator compartment cold air supply connecting portion **310**, a hook **314** may be formed on the duct bottom cover **305** to be guided and secured to the hook securing portion **211** of the connection duct **200**.
(154) A connection duct securing member **213** may protrude from the upper surface of the connection duct **200** in a direction to the front surface of the refrigerator **51**.
(155) A fastening member through hole **313** through which a fastening member **311** such as a bolt passes to be fastened to the connection duct securing member **213** may be formed on the lower surface of the refrigerator compartment cold air supply connecting portion **310**.
(156) As the hook securing portion **211** is formed on the upper surface of the connection duct **200**, it may be possible to guide an accurate coupling position in the process of coupling connection duct to the refrigerator compartment cold air supply duct **300**.
(157) Since the connection duct securing member **213** is formed on the upper surface of the connection duct **200**, the stronger securing force with the refrigerator compartment cold air supply duct **300** may be provided.
(158) Accordingly, the upper region of the refrigerator cold air supply duct **300** according to the embodiment of the present disclosure may have some area inserted in the duct inserting groove **49** even without separate fastening means, thereby strongly fixing the upper region.
(159) The lower region of the refrigerator compartment cold air supply duct **300** may be strongly fixed by using only the fastening member **311** such as a bolt. Accordingly, the workability of workers who assemble in the form of module assembly may be greatly improved.
(160) The refrigerator **1** according to the present disclosure may include a refrigerator compartment cold air returning duct configured to return and supply the cold air of the refrigerator compartment to the evaporator **101**.
(161) The refrigerator compartment cold air returning duct **500** may have one end connected to the freezer compartment **52** and the other end connected to the refrigerator compartment **51**. The both ends of the refrigerator compartment cold air returning duct **500** may overlap with each other in a vertical direction.
(162) One end of the refrigerator compartment cold air returning duct **500** may be configured to communicate with the freezer compartment **52** through a refrigerator compartment returning communication outlet hole **46***b*.
(163) The refrigerator compartment cold air returning duct **500** may pass through the rear surface of the evaporator **101**.
(164) If the refrigerator compartment cold air returning duct **500** is disposed too close to the evaporator **101**, dew condensation could occur. Accordingly, the refrigerator compartment cold air returning duct **500** may be spaced apart a preset distance apart from the evaporator **101**.
(165) In particular, since the refrigerator compartment cold air returning duct **500** may have a cold air flow path, frost could be more likely to occur.
(166) Accordingly, one end and the other end of the refrigerator compartment cold air returning duct **500** may be disposed to overlap with each other in the vertical direction in order to have a cold air flow path as short as possible.
(167) In addition, both lateral surfaces of the refrigerator compartment cold air returning duct **500** may be formed in a straight line shape parallel to each other by reducing curvature as much as

possible. Accordingly, the refrigerator compartment cold air returning duct **500** may allow the cold air flow path corresponding to the area overlapping the evaporator **101** to have the shortest distance.
(168) Accordingly, the refrigerator compartment cold air returning duct **500** may the shortest length of the area overlapping the evaporator **101**, so that the possibility of frost may be reduced.
(169) In addition, the refrigerator compartment cold air returning duct **500** can uniformly form frost without biasing the cold air passing through the refrigerator compartment cold air returning duct, or can reduce the formation of frost.
(170) One end and the other end of the refrigerator compartment cold air returning duct **500** may be disposed to pass through the center of the refrigerator compartment **51** and the center of the freezer compartment **52** with respect to the left-right direction.
(171) A refrigerator compartment cold air returning communication hole **61** in communication with the other end of the refrigerator compartment cold air returning duct **500** may be provided on the lower surface **41***c* of the refrigerating case. The refrigerator compartment cold air returning communication hole **61** may be disposed to pass through the center **1***c* of the refrigerator compartment **51**.
(172) It is possible to balance the overall cold air of the refrigerator **1**, because one end and the other end of the refrigerator compartment cold air returning duct **500** are disposed to pass through the center of the refrigerator compartment **51** and the center of the freezer compartment **52** with respect to the left-right direction, thereby balancing the overall cold air inside the refrigerator **1**.
(173) The refrigerator compartment returning duct **500** may be configured to return the cold air circulating after supplied to the refrigerator compartment **51** to the freezer compartment **52**.
(174) The other end of the refrigerator compartment cold air returning duct **500** in communication with the refrigerator compartment cold air returning communication hole **61** for returning the cold air from the refrigerator compartment **51** may be disposed to pass through the center of the refrigerator compartment **51**, thereby inducing the cold air returning pass flow as naturally as possible.
(175) Referring to FIGS. **23**A and **23**B additionally, a returning duct rear extended portion **521** may be formed at the other end of the refrigerator compartment cold air returning duct **500**. The returning duct rear extended portion **521** may be vertically extended along a rear end of the upper surface of the refrigerator compartment cold air returning duct **500**.
(176) In addition, a returning duct side extended portion **522** may be formed at the other end of the refrigerator compartment cold air returning duct **500**. The returning duct side extended portion **522** may be vertically extended along one side end of the upper surface of the refrigerator compartment cold air returning duct **500**.
(177) The returning duct rear extended portion **521** and the returning duct side extended portion **522** may be connected to each other.
(178) Accordingly, the other end of the refrigerator compartment cold air returning duct **500** may be fixed to the refrigerating case **41** to support the lower surface **41***c* and the rear surface **41***a* of the refrigerating case.
(179) Specifically, the other end of the refrigerator compartment cold air returning duct **500** may be configured to support the lower surface **41***c* of the refrigerating case, and the returning duct rear extended portion **521** and the returning duct side extended portion **522** may be configured to support the rear surface of the refrigerating case **41**.
(180) In this instance, the returning duct side extended portion **522** may be configured to partially support a rear surface of the duct inserting groove **49**.
(181) The refrigerator compartment cold air returning duct **500** according to the embodiment of the present disclosure may serve not only to return the cold air of the refrigerator compartment **51** but also serve as a support member capable of strongly supporting the refrigerating case **41**.
(182) The connection duct **200** described above may be disposed between the refrigerator

compartment cold air returning duct **500** and the lateral surface **41***d* of the refrigerating case that is one lateral surface of the refrigerator compartment **51** with respect to the left-right direction.
(183) In the connection duct **200** having one end connected to the freezer compartment **52** and the other end connected to the refrigerator compartment **51**, the width of the cold air flow path may increase from one end to the other end of the connection duct **200**.
(184) In this instance. The width of the cold air flow path of the connection duct **200** may increase in a direction toward the center of the refrigerating compartment **51**.
(185) The other end of the connection duct **200** may be disposed to pass through the center of the refrigerator compartment **51** in the left-right direction.
(186) As mentioned above, the connection duct **200** may be disposed between the refrigerator compartment cold air returning duct **500** and one lateral surface of the refrigerator compartment **51**, and the width of the cold air flow path may increase in the direction toward the center o the refrigerator compartment **51**, so that it may be possible to balance the overall cold air of the refrigerator **1**.
(187) In addition, since the other end of the connection duct **200** is in communication with the refrigerator compartment cold air supply duct **300**, the connection duct **200** and the refrigerator compartment cold air supply duct **300** may communicate with each other at the center of the refrigerator compartment **51** as much as possible.
(188) The refrigerator compartment **51** may be divided into a first storage chamber Ma and a second storage chamber **51***b.*
(189) A second storage chamber cold air supply duct **400** may be configured to supply cold air to the second storage chamber **51***b*, and the second storage cold air supply duct **400** may be disposed on the outer surface of the refrigerating case **41**.
(190) The heat insulating material **11** may be foamed and filled in the space between the inner case **40** and the outer case **10**.
(191) The second storage chamber cold air supply duct **400** may be disposed to pass through the space foamed and filled with the insulating material **11**, to be embedded in the space between the inner case **40** and the outer case **10**.
(192) The second storage chamber cold air supply duct **400** disposed on the outside of the refrigerating case **41** may not be exposed when the upper door **20** of the refrigerator **1** is opened.
(193) As described above, the rear projected portion **43** projected inward of the refrigerating case **41** may be provided on the rear surface **41***a* of the refrigerating case in order to receive at least predetermined area of the connection duct **200** from the outside o the refrigerating case **41**.
(194) In this instance, the rear projected portion **43** may be formed in a shape capable of receiving the at least predetermined area of the second storage chamber cold air supply duct **400** in addition to the shape capable of receiving the connection duct **200**.
(195) The connection duct **200** and the second storage chamber cold air supply duct **400** may be disposed adjacent to each other.
(196) For example, the second storage chamber cold air supply duct **400** may be disposed between the connection duct **200** and the lateral surface **41***d* of the refrigerating case that is one surface of the refrigerator compartment **51**.
(197) Since the second storage chamber cold air supply duct **400** is configured to supply cold air to the second storage chamber **51***b*, with the shape embedded in the outer surface of the refrigerating case **41**, the area projected toward the inside of the refrigerator compartment **51** may be reduced and the inner volume of the refrigerator compartment **51** may be then increased.
(198) In addition, since the second storage chamber cold air supply duct **400** is disposed to pass through the space between the inner case **40** and the outer case **10**, which is foamed with the insulating material **11**, there is no need of providing an additional insulating material for preventing heat exchange between the second storage cold air supply duct **400** through which the cold air passes.

(199) According to the embodiment of the present disclosure, sufficient insulation effect may be achieved by using the insulating material **11** foamed
(200) According to the embodiment of the present disclosure, a separate heat insulating member for imparting an insulating effect to the second storage chamber cold air supply duct **400** may not additionally required so that the thickness of the second storage chamber cold air supply duct **400** may be greatly reduced.
(201) Accordingly, the thickness of the rear projected portion **43** projected toward the inside of the refrigerating case **41** may be greatly reduced so that the inner volume of the refrigerator compartment **51** may be increased.
(202) Referring to FIG. **11**, a second storage chamber cold air outlet cover **440** may be provided on a front surface of the other end of the second storage chamber cold air supply duct **400** connected to the second storage chamber **51***b*.
(203) The second storage chamber cold air outlet cover **440** may be disposed inside the refrigerating case **41**.
(204) The other end of the second storage chamber cold air supply duct **400** may have a flat plate shape directed toward the front surface of the refrigerator compartment **51** to communicate with the second storage chamber cold air supply communication hole **45** formed in the rear surface **41***a* of the refrigerating case.
(205) Specifically, the second storage chamber cold air supply outlet **400***b* may be formed in the other end of the second storage chamber cold air supply duct **400** to be in communication with the second storage chamber cold air supply communication hole **45** formed in the rear surface **41***a* of the refrigerating case.
(206) A temperature sensor inserting portion **409** for receiving a temperature sensor may be formed in the other end of the second storage chamber cold air supply duct **400** and configured to sense the temperature of the second storage chamber **51***b* in order to adjust cold air supply of cold air to the second storage chamber **51***b*.
(207) In addition, a pair of second storage chamber cover coupling members may be formed in the other end of the second storage cold air supply duct **400** to facilitate the coupling and decoupling of the second storage chamber cold air outlet cover **440** disposed on the front surface.
(208) The second storage chamber cover coupling member **407** may have a shape that may be hooked to the second storage chamber cold air outlet cover **440**.
(209) The second storage cold air outlet cover **440** may include a second storage chamber temperature sensor **443** configured to sense the temperature of the second storage chamber **51***b* by exposing the temperature sensor.
(210) The second storage chamber cold air outlet cover **440** may include a second storage chamber cold air outlet guide **441** configured to guide a direction of discharging the cold air supplied to the second storage chamber **51***b*.
(211) The second storage chamber cold air supply duct **400** may be disposed adjacent to the lateral surface **41***d* of the refrigerating case with respect to the center of the refrigerator compartment **51**. Due to that, the cold air of the second storage chamber may be discharged in a state of being biased toward the lateral surface **41***d* of the refrigerating case.
(212) Accordingly, the second storage chamber cold air outlet guide **441** may have a plurality of guide ribs **442** formed to guide the cold air in a direction as much as possible toward the other lateral surface **41***e* of the refrigerating case in order to guide the cold air toward the other lateral surface **41***e* of the refrigerating case.
(213) Accordingly, the cold air inside the second storage chamber **51***b* may be circulated inside the second storage chamber **51***b* as uniformly as possible.
(214) The second storage chamber cold air supply duct **400** may be embedded in the space between the inner case **40** and the outer case **10**, which is foamed with the insulating material **11**.
(215) Accordingly, since only the second storage chamber cold air outlet cover **440** is disposed

inside the inner case **40**, the projected area inside the refrigerator compartment **51** for accommodating components related to the second storage chamber cold air supply system may be reduced only to increase the inner volume of the refrigerator compartment **51**.
(216) The refrigerator **1** according to the present disclosure may include the ice-making chamber **22** provided in the upper door **20** configured to open and the close the refrigerator compartment **51**.
(217) The cold air generated by the evaporator **101** may be supplied to the ice-making chamber **22** through an ice-making chamber cold air supply duct **600**.
(218) An ice-making chamber cold air supply inlet hole **600***a* may be formed in one end of the ice-making chamber cold air supply duct **600** to be in communication with the grill fan assembly **100** through an ice-making chamber cold air supply communication inlet hole **47***a* of the freezing case **42**.
(219) In this instance, an ice-making cold air guide duct **610** may be disposed between the ice-making chamber cold air supply duct **600** and the grill fan assembly **100** to facilitate communication between the cold air supply duct **600** and the frill fan assembly **100**.
(220) The ice-making chamber cold air supply duct **610** may be configured to switch a direction of the cold air discharged from the frill fan assembly **100**.
(221) For example, the ice-making chamber cold air supply communication inlet hole **47***a* for discharging the cold air from the grill fan assembly **100** may be formed as a through-hole formed upward to have an inclined surface toward the other lateral surface **42***e* of the freezing case.
(222) The ice-making chamber cold air guide duct **610** may be in communication with the ice-making chamber cold air supply communication inlet hole **47***a* to switch a flow direction of the discharged cold air.
(223) The other end of the ice-making chamber cold air supply duct **600** may be in communication with the ice-making chamber **22** through the ice-making chamber cold air supply communication outlet hole **47***b* formed on the other surface **41***e* of the refrigerating case.
(224) The cold air circulated in the ice-making chamber **22** may return to the freezer compartment **52** through an ice-making chamber cold air returning duct **700**.
(225) An ice-making chamber cold air returning outlet hole **700***b* may be formed in one end of the ice-making chamber cold air returning duct **700** to communicate with the freezer compartment **52** through an ice-making chamber cold air returning communication outlet hole **48***b* formed on the other lateral surface **42***e* of the freezing case.
(226) An ice-making returning inlet hole **700***a* may be formed in the other end of the ice-making chamber cold air returning duct **700** to communicate with the ice-making chamber **22** through an ice-making chamber cold air returning communication inlet hole **48***a* formed on the other lateral surface **42***e* of the freezing case.
(227) As another example, when the ice-making chamber **22** is disposed in the second upper door **20***b*, the ice-making chamber cold air returning communication outlet hole **48***b* and the ice-making chamber cold air returning communication inlet hole **48***a* may be disposed on one lateral surface **42***d* of the freezing case.
(228) As described above, the upper door **20** including the ice-making chamber **22** may be positioned on the front surface of the refrigerator **1**.
(229) Accordingly, the ice-making chamber cold air supply duct **600** and the ice-making chamber cold air returning duct **700** may be extended along the other lateral surface **41***e* of the refrigerating case to be facilitate communication between the ice-making chamber **22** and a cold air path of the refrigerating case **42**.
(230) In this instance, the other lateral surface **41***e* of the refrigerating case through which the ice-making chamber cold air supply duct **600** and the ice-making chamber cold air returning duct **700** pass may have a relatively low temperature, considering the overall temperature distribution of the refrigerator **1**.
(231) Accordingly, the lateral surface **41***d* and the other lateral surface **41***e* of the refrigerating case

may have cold air imbalance.
(232) To reduce cold air imbalance in a left-right direction, the connection duct **200** and the second storage chamber cold air supply duct **400** may be disposed adjacent to the surface **41***d* that faces the other lateral surface **41***e* of the refrigerating case, to balance cold air of the overall refrigerator **1**.
(233) In this regard, the refrigerator compartment cold air supply duct **300** may include a first refrigerator compartment cold air flow path **321** and a second refrigerator compartment cold air flow path **322** that are configured to branch the cold air guided from the connection duct **200**.
(234) The first refrigerator compartment cold air flow path **321** may be formed to have a wider cold air flow path width than the second refrigerator compartment cold air flow path **322**, so as to induce more cold air toward the first refrigerator compartment cold air path **321**.
(235) The second refrigerator compartment cold air flow path **322** through which relatively less cold air is induced may be disposed closer to the other lateral surface **41***e* of the refrigerating case, on which the ice-making chamber cold air supply duct **600** is disposed, to the first refrigerator compartment cold air flow path **321**.
(236) As described above, the refrigerator compartment cold air supply duct **300** may have the first refrigerator compartment cold air flow path **321** having a larger cold air flow path width that is disposed father from the other lateral surface **41***e* of the refrigerating case where the ice-making chamber cold air supply duct **600** than the second refrigerator compartment cold air flow path **322**, so that the overall cold air of the refrigerator **1** may be balanced.
(237) Meanwhile, a flow path opening/closing module **130** for selectively cut off the supply of the cold air generated by the evaporator **101** to the refrigerator compartment **51** may be disposed in the freezer compartment **52**.
(238) Referring to FIG. **24**A, the flow path opening/closing module **130** may include a first storage chamber flow path opening/closing damper **140**, and may be configured to selectively cut off the cold air supplied to the first storage chamber **51***a* through the connection duct **200** and the refrigerator compartment cold air supply duct **300**.
(239) Unless the refrigerator **51** is divided into a plurality of spaces in which the temperatures are separately adjusted, the flow path opening/closing module **130** may include only the first storage chamber flow path opening/closing damper **140** and the cold air of the refrigerator compartment **51** may be supplied through the refrigerator compartment cold air supply duct **300**.
(240) As another example, when the refrigerator compartment **51** includes the first storage chamber **51***a* and the second storage chamber **51***b* that are adjusted to have different temperatures, respectively, the flow path opening/closing module **130** may include a first storage chamber flow path opening/closing damper **140** and a second storage chamber flow path opening/closing damper **150**.
(241) The first storage chamber flow path opening/closing damper **140** may be configured to selectively cut off the cold air supplied to the first storage chamber **51***a* through the connection duct **200** and the refrigerator compartment cold air supply duct **300**.
(242) The second storage chamber opening/closing damper **150** may be configured to selectively cut off the cold air supplied to the second storage chamber **51***b* through the second storage chamber cold air supply duct **400**.
(243) The first storage chamber flow path opening/closing damper **140** and the second storage chamber flow path opening/closing damper **150** for selectively cutting off the cold air supply to the first storage chamber **51***a* and the second storage chamber **51***b*, respectively, may be provided not in the refrigerator compartment **51** but in the freezer compartment **52**.
(244) If the flow path opening/closing dampers **140** and **150** are provided in the refrigerator compartment **51**, the refrigerator compartment **51** could have a shape more projected inward as much as the area occupied by the first and second storage chamber flow path opening/closing dampers **140** and **150**. Due to this structure, the inner volume of the refrigerator compartment **51** may be reduced.

(245) However, since the flow path opening/closing module **130** including the flow path opening/closing dampers **140** and **150** are disposed in the freezer compartment **52**, not the refrigerator compartment **51**, the area of the projected part projected to the inside of the refrigerator compartment **51** may be reduced only to increase the inner volume of the refrigerator compartment **51**.
(246) In the refrigerator **1** according to the present disclosure, the flow path opening/closing module **130** including the flow path opening/closing dampers **140** and **150** may be disposed in the freezer compartment **52**, not the refrigerator compartment **51**. Accordingly, even when the flow path opening/closing dampers **140** and **150** are closed, the cold air of the freezer compartment **52** may fail to rise to the refrigerator compartment but stay in the freezer compartment **52**.
(247) Accordingly, the refrigerator **1** according to the present disclosure may greatly reduce dew condensation near the flow path opening/closing dampers **140** and **150**.
(248) As described above, the refrigerator **1** according to the present disclosure may have cold air circulation flow below.
(249) Referring to FIG. **14**, cold air supply flow to the first storage chamber **51***a* and cold air returning flow after circulating in the first storage chamber **51***a* will be described.
(250) The cold air generated by the evaporator **101** disposed in the freezer compartment **52** may be blown to the connection duct **200** embedded in the rear outer surface of the refrigerator compartment **51** by a grill fan assembly **100** disposed in the freezer compartment **52**.
(251) The cold air blown to the connection duct **200** may communicate with the refrigerator compartment cold air supply duct **300** disposed on the rear surface of the refrigerator compartment **51** to be guided to the refrigerator compartment cold air supply duct **300**.
(252) The refrigerator compartment cold air supply duct **300** may be configured to discharge cold air forward from the upper region of the refrigerator compartment **51**.
(253) The cold air discharged forward from the upper region of the refrigerator compartment **51** may circulate inside the refrigerator compartment **51** and return to the rear surface in the lower region of the refrigerator compartment **51**.
(254) Since the refrigerator compartment cold air returning duct **500** is in communication with the lower region of the rear surface in the refrigerator compartment **51**, the cold air circulated in the refrigerator compartment **51** may return to the freezer compartment **52** through the refrigerator compartment cold air returning duct **500**.
(255) Referring to FIG. **15**, cold air flow to the second storage chamber **51***b* of the refrigerator compartment **51** and cold air returning flow after circulating inside the second storage chamber **51***b* will be described.
(256) The cold air generated by the evaporator **101** disposed in the freezer compartment **52** may be blown to the connection duct **200** embedded in the rear outer surface of the refrigerator compartment **51** by the grill fan assembly **100** disposed in the freezer compartment **52**.
(257) The cold air blown to the second storage chamber cold air supply duct **400** may discharge cold air to the second storage chamber **51***b* from the rear surface of the refrigerator compartment **51**.
(258) The second storage chamber cold air supply duct **400** may discharge cold air forward from the upper region of the second storage chamber **51***b*.
(259) The cold air discharged forward from the upper region of the second storage chamber **51***b* may circulate inside the second storage chamber **51***b* and return to the rear surface in the lower region of the second storage chamber **51***b*.
(260) Since the refrigerator compartment cold air returning duct **500** is in communication with the lower region of the rear surface in the second storage chamber **51***b*, the cold air circulated in the second storage chamber **51** may return to the freezer compartment **52** through the refrigerator compartment cold air returning duct **500**.
(261) As described above, the cold air ducts for supplying cold air to the first storage chamber **51***a*

and the second storage chamber **51**_b_, respectively, are different from each other. However, the cold air circulating the first storage chamber **51**_a_ and the second storage chamber **51**_b_ may return through the same refrigerator compartment cold air returning duct **500**.
(262) Referring to FIG. **16**, cold air supply flow to the freezer compartment **52** and cold air returning flow after circulating the freezer compartment **52** will be described.
(263) The cold air generated by the evaporator **101** disposed in the freezer compartment **52** may be blown to the connection duct **200** embedded in the rear outer surface of the refrigerator compartment **51** by the grill fan assembly **100** disposed in the freezer compartment **52**.
(264) The grill fan assembly **00** may discharge cold air forward from the upper region of the freezer compartment **52**.
(265) The cold air discharged forward from the upper region of the freezer compartment **52** may circulate inside the freezer compartment **52** and return to the rear surface in the lower region of the freezer compartment **52**.
(266) Since the mechanical chamber **53** is provided in a lower rear area of the freezer compartment **52**, a lower rear surface of the freezer compartment **52** may have an inclined surface that rises obliquely from a lower area.
(267) The cold air returning to the rear surface of the freezer compartment **52** may flow into the freezer compartment cold air returning guide **119** of the grill fan assembly **100** along the inclined surface of the rear surface in the lower region of the freezer compartment **52**.
(268) Referring to FIG. **17**, cold air supply flow to the ice-making chamber **22** and cold air returning flow after circulating inside the ice-making chamber **22** will be described.
(269) The cold air generated by the evaporator **101** may be supplied to the ice-making chamber **22** disposed in the first upper door **20**_a_ coupled to the front surface of the refrigerator **1** through the ice-making chamber cold air supply duct **600**.
(270) One end of the ice-making chamber cold air supply duct **600** may be in communication with the grill fan assembly **100** through the ice-making chamber cold air supply communication inlet hole **47**_a_ of the freezing case **42**.
(271) The other end of the ice-making chamber cold air supply duct **600** may be in communication with the ice-making chamber cold air inlet hole **22**_a_ of the ice-making chamber **22** through the ice-making chamber cold air supply communication outlet hole **47**_b_ provided on the other lateral surface **41**_e_ of the refrigerating case.
(272) In addition, the cold air circulated inside the ice-making chamber **22** may returned to the freezer compartment **52** through the ice-making chamber cold air returning duct **700**.
(273) One end of the ice-making chamber cold air returning duct **700** may be in communication with the freezer compartment **52** through the ice-making chamber cold air returning communication outlet hole **48**_b_ disposed on the other lateral surface **42**_e_ of the freezing case.
(274) The other end of the ice-making chamber cold air returning duct **700** may be in communication with the ice-making chamber cold air returning hole **22**_b_ of the ice-making chamber **22** through the ice-making chamber cold air returning communication inlet hole **48**_a_ disposed on the other lateral surface **42**_e_ of the freezing case.
(275) The cold air returning to the freezer compartment **52** from the ice-making chamber **22** may be guided by the freezer compartment cold air returning guide **119** disposed in a lower area of the grill fan assembly **100** of the freezer compartment **52**.
(276) As described above, the cold air supply ducts for supplying cold air to the freezer compartment **52** and the ice-making chamber **22** are different from each other. However, the cold air circulating inside the freezer compartment **52** and the ice-making chamber **22** may return through the same freezer compartment cold air returning guide **119**.
(277) [Refrigerator Compartment Cold Air Supply Duct]
(278) Hereinafter, referring to FIGS. **18** to **21**, the refrigerator compartment cold air supply duct **300** will be described in detail.

(279) The refrigerator compartment cold air supply duct **300** may include a duct body defining a front surface, a duct sheet **304** spaced apart behind the duct body **302**, and a duct insulating portion **303** provided between the duct body **302** and the duct sheet **304** to form a first refrigerating chamber cold air flow path **321** and a second refrigerating chamber air flow **322**.
(280) A decorative panel **301** may be provided on a front surface of the duct body **302**.
(281) The decorative panel **301** may define an exterior design of a rear wall surface (or a rear lateral surface) of the first storage chamber **51***a*, and may be formed of metal (e.g., stainless steel).
(282) A shelf securing portion **309** may be formed in a center area of the decorative panel **301** in a height direction.
(283) Through holes corresponding to the shelf securing portion **309** may be formed in the duct body **302**, the duct insulating portion **303** and the duct sheet **304**, respectively, which are disposed on the rear surface of the decorative panel **301**.
(284) The duct body **302** may be formed in the same size and shape as the rear surface **41***a* of the refrigerating.
(285) However, the duct body **302** may be formed narrower than the width of the rear surface **41***a* of the refrigerating case in the left-right direction to have a predetermined space spaced apart a preset distance from one lateral surface **41***d* and the other lateral surface **41***e* of the refrigerating case.
(286) Accordingly, the duct body **302** may not be in contact with the lateral surface **41***d* and the other lateral surface **41***e* of the refrigerating case.
(287) A pair of lighting units **360** may be disposed in both sides of a rear surface of the duct body **302** along a longitudinal direction of the duct body **302**, respectively.
(288) The pair of the lighting units **360** may be disposed to face the lateral surface **41***d* and the other lateral surface **41***e* of the refrigerating case to irradiate light toward the lateral surface **41***d* and the other lateral surface **41***e* of the refrigerating case.
(289) The lighting units **360** may be fixedly coupled to the duct body **302** by using a lighting unit fixing member **378** and a lighting unit coupling member **379** (e.g., a bolt) that are disposed on both sides of the rear surface of the duct body **302**.
(290) An upper area of the refrigerator compartment cold air supply duct **300** may be inserted in a duct inserting groove **49** formed in the upper surface **41***b* of the refrigerating case to be fixed to the refrigerating case **41**.
(291) A refrigerator compartment cold air supply connecting portion **310** may be formed in a lower area of the duct body **302**.
(292) Referring to FIGS. **21**A and **21**B, the refrigerator compartment cold air supply connecting portion **310** of the refrigerator compartment cold air supply duct **300** may be connected to the connection duct **200** disposed below.
(293) The refrigerator compartment cold air supply connecting portion **310** may have a pair of coupling member through holes **313**, and a connection duct securing member **213** may be formed on the upper surface of the connection duct **200**.
(294) The coupling member **311** such as a bolt may be coupled to the connection duct securing member **213** through the coupling member through hole **313**, to fixedly coupled the lower area of the refrigerator compartment cold air supply duct **300** to the connection duct **200**.
(295) A duct lower cover **305** may be disposed on a lower surface of the refrigerator compartment cold air supply connecting portion **310**, and the refrigerator compartment cold air supply connecting portion **310** and the connection duct **200** may be in communication while having the duct lower cover **305** disposed therebetween.
(296) A duct lower gasket **306** may be disposed on the lower surface of the duct lower cover **305** to improve the airtightness between the refrigerator compartment cold air supply connecting portion **310** and the connection duct **200**.
(297) A plurality of securing bosses **398** may be formed on a rear surface of the duct body **302**.

(298) Coupling through-holes **60** corresponding to the securing bosses **398** may be formed on the rear surface **41***a* of the refrigerating case.
(299) A separate coupling bush **399** may be provided on the outer surface of the refrigerating case **41** to be coupled to the securing boss **398** through the coupling through hole **60**.
(300) Accordingly, the refrigerator compartment cold air supply duct **300** may be secured to the rear surface **41***a* of the refrigerating case by coupling the securing bosses **398** to the coupling bush **399** to have a strong coupling force.
(301) A duct insulating portion **303** having a predetermined thickness may be disposed in a rear area of the duct body **302**.
(302) The duct insulating portion **303** may have a cold air flow path formed therein to form a protrusion pattern backward with respect to the duct insulating portion **303**.
(303) The duct sheet **304** having a shape corresponding to the cold air flow path of the duct insulating portion **303** to be coupled to the duct insulating portion **303** may be disposed on a rear area of the duct insulating portion **303** to airtight cover the cold air flow path formed in the duct insulating portion **303**.
(304) For example, the duct sheet **304** may be formed of a heat insulating material to reduce heat loss of the cold air flowing along the inside of the first refrigerating chamber cold air flow path **321** and the inside of the second refrigerating chamber cold air flow path **322**.
(305) The duct sheet **304** may have a thickness thin enough to airtight seal the first refrigerating chamber cold air flow path **321** and the second refrigerating chamber cold air flow path **322**, so that the overall thickness of the refrigerator compartment cold air supply duct **300**.
(306) However, when the duct sheet **304** is formed with the thickness that is as thin as possible, the supporting force of the duct sheet **304** could be insufficient.
(307) Accordingly, a plurality of cold air guide ribs **323** may be formed in a backward direction of the duct insulating portion **303** to support the duct sheet by inducing the cold air flow direction.
(308) Due to the coupling structure between the duct insulating portion **303** and the duct sheet **304**, the first refrigerating chamber cold air flow path **321** and the second refrigerating chamber cold air flow path **322** may be formed in the refrigerator compartment cold air supply duct **300**.
(309) The first refrigerating chamber cold air flow path **321** and the second refrigerating chamber cold air flow path **322** may be branched from a lower area of the duct insulating portion **303**.
(310) The cold air passing through the first refrigerating chamber cold air flow path **321** and the second refrigerating chamber cold air flow path **322** may be discharged from an upper area of the duct insulating portion **303** to the refrigerator compartment **51** through a first refrigerating chamber cold air main outlet hole **341** and a second refrigerating chamber cold air main outlet hole **342**.
(311) A refrigerator compartment cold air guide **324** may be formed on an upper end of the duct insulating portion **303** to uniformly discharge the cold air from the first refrigerator compartment cold air main outlet hole **342** and the second refrigerator compartment cold air main outlet hole **342**.
(312) The refrigerator compartment cold air guide **324** may be formed on a rear surface of the duct insulating portion **303** as an island shape, may have the same height as cold guide rib **323**.
(313) A refrigerator compartment cold air main outlet guide **349** may be formed in an upper end of the duct body **302** and configured to guide the cold air discharged from the second refrigerator compartment cold air main outlet hole **342** and the second refrigerator compartment cold air main outlet hole **342** to be discharged toward the front surface of the refrigerator compartment **51**.
(314) The refrigerator compartment cold air main outlet guide **349** may be configured to guide the cold air discharged to the refrigerator compartment **51** toward the front surface.
(315) The first refrigerator compartment cold air flow path **321** disposed adjacent to the lateral surface **41***d* of the refrigerating case may have a greater cold air flow path width than the second refrigerator compartment cold air flow path **322** disposed adjacent to the other lateral surface **41***e* of the refrigerating case.

(316) Accordingly, even when the ice-making chamber cold air supply duct **600** and the ice-making chamber cold air returning duct **700** are disposed along the other lateral surface **41***e* of the refrigerating case, more cold air may be sent toward the first refrigerator compartment cold air flow path **321** so that the overall cold air of the refrigerator **1** may be balanced.
(317) The duct insulating portion **303** may include one or more refrigerator compartment cold air auxiliary flow path **325** branched from the first refrigerator compartment cold air flow path **321** and/or the second refrigerator compartment cold air flow path **322**.
(318) For example, the refrigerator compartment cold air auxiliary flow path **325** may be additionally branched from an upper area of the first refrigerator compartment cold air flow path **321** and/or the second refrigerator compartment cold air flow path **322** in a direction to the center of the duct insulating portion **303**.
(319) In an embodiment of the present disclosure, auxiliary cold air flow paths may be branched from the first refrigerator compartment cold air flow path **321** and the second refrigerator compartment cold air flow path **322**, respectively, to form a pair of refrigerator compartment cold air auxiliary flow paths **235**.
(320) The duct insulating portion **303** may include a refrigerator compartment cold air auxiliary outlet hole **330** configured to discharge the cold air guided through the refrigerator compartment cold air auxiliary flow path **325**.
(321) The refrigerator compartment cold air auxiliary flow paths **325** may be branched from an upper area of the first refrigerator compartment cold air flow path **321** and an upper area of the second refrigerator compartment cold air flow path **322** with respect to the center of the refrigerator compartment **41** in a direction to the center of the refrigerator compartment **51**.
(322) Accordingly, the refrigerator compartment cold air auxiliary outlet hole **330** may be provided in an upper area with respect to the center of the refrigerator compartment **51**.
(323) The refrigerator compartment cold air auxiliary outlet hole **330** may be disposed in a lower area than the first refrigerator compartment cold air main outlet hole **341** and the second refrigerator compartment cold air main outlet hole **342**.
(324) A refrigerator compartment cold air auxiliary outlet guide **339** including a refrigerator compartment filter **331** may be disposed on a front surface of the refrigerator compartment cold air auxiliary outlet hole **330**.
(325) The cold air supplied from the refrigerator compartment cold air auxiliary outlet hole **330** may pass through the refrigerator compartment filter **331**.
(326) The refrigerator compartment cold air auxiliary outlet hole **330** may include a first refrigerator compartment cold air auxiliary outlet hole **330***a* and a second refrigerator compartment cold air auxiliary outlet hole **330***b.*
(327) The refrigerator compartment filter **331** may be a deodorizing filter including activated carbon that removes odors.
(328) Accordingly, the refrigerator **1** according to the present disclosure does not need to include a separate suction fan provided in the refrigerator compartment **51** in order to deodorize the refrigerator compartment **51**.
(329) The area projected toward the inside of the refrigerator compartment **51** by the suction fan may be reduced, thereby reducing the inner volume of the refrigerator compartment **51**.
(330) The refrigerator **1** according to the present disclosure may include a refrigerator compartment cold air auxiliary flow path **325** configured to discharge cold air instead of using the suction fan for sucking cold air from the refrigerator compartment **51**.
(331) Accordingly, the deodorizing function for the refrigerator compartment may be performed without disturbing the circulation structure of the cold air supplied and circulated inside the refrigerator compartment.
(332) In addition, the refrigerator **1** according to the present disclosure may perform the deodorizing function by using the cold air supply circulation for continuously circulating cold air.

(333) Accordingly, in preparation for performing the deodorizing function by intermittently operating the suction fan, the total amount of cold air passing through the refrigerator compartment filter **331** may be similar or increased to improve the deodorizing function.
(334) The refrigerator compartment cold air auxiliary outlet guide **339** may further include an insulating member **332** disposed on a front surface of the refrigerator compartment filter **331**.
(335) The insulating member **332** may be spaced apart from the refrigerator compartment filter **331**, so that a cold air flow path through which cold air passes may be formed between the insulating member **332** and the refrigerator compartment filter **331**.
(336) The refrigerator compartment cold air auxiliary outlet guide **339** may be configured to guide the cold air having passed through the refrigerator compartment filter **331** to be discharged downward inside the refrigerator compartment **51** through the space between the insulating member **332** and the refrigerator compartment filter **331**.
(337) Accordingly, the refrigerator compartment cold air auxiliary outlet guide **339** may have s shape covering the front surface of the refrigerator compartment cold air auxiliary outlet hole **330**, and may have a guide cold air outlet hole **351** formed on a lower surface to discharge cold air.
(338) The cold air discharged through the refrigerator compartment cold air auxiliary outlet hole **330** may not be directly discharged to the front surface of the refrigerator compartment **51**, but the cold air may have a flow path direction changed from the front surface to the bottom by the refrigerator compartment cold air auxiliary outlet guide **339**.
(339) Accordingly, the cold air discharged through the refrigerator compartment cold air auxiliary outlet hole **330** may primarily collide with the refrigerator compartment cold air auxiliary outlet guide **339**.
(340) When the cold air collides with the refrigerator compartment cold air auxiliary outlet guide **339** having the surface exposed to the relatively humid refrigerator compartment **51**, dew condensation might occur on the refrigerator compartment cold air auxiliary outlet guide **339**.
(341) According to the present disclosure, the refrigerator compartment cold air auxiliary outlet guide **339** may reduce occurrence of dew condensation, because the insulating material **332** is disposed on the front surface of the refrigerator compartment filter **331**.
(342) The refrigerator compartment cold air auxiliary outlet guide **339** described above may overlap with the refrigerator compartment cold air returning duct **500** disposed in the lower area of the refrigerator compartment **51** in a vertical direction.
(343) Specifically, the refrigerator compartment cold air auxiliary outlet hole **330** of the refrigerator compartment cold air supply duct **300** may vertically overlap with one end and the other end of the refrigerator compartment cold air returning duct **500** disposed in the lower area of the refrigerator compartment **51**.
(344) Accordingly, the present disclosure may induce a cold air flow path that allows the cold air deodorized while passing through the refrigerator compartment filter **331** to easily escape to the refrigerator compartment cold air returning duct **500**, without interfering with the flow of the cold air circulating inside the refrigerator compartment **51** as much as possible.
(345) In addition, since the cold air discharged from the upper region of the refrigerator compartment **51** by the refrigerator compartment cold air main outlet guide **349** is directed to the front surface, the cold air discharged from the refrigerator compartment cold air main outlet guide **349** could be difficult to circulate toward the rear surface **41***a* of the refrigerating case.
(346) According to the present disclosure, the cold air discharged through the refrigerator compartment cold air auxiliary outlet hole **330** may circulate toward the rear surface **41** of the refrigerating case. Due to this structure, the overall cold air supply of the refrigerator compartment **51** may be performed more uniformly and smoothly.
(347) In addition, the duct body **302** may include a sterilization unit **333** including an ultraviolet sterilizing device.
(348) The ultraviolet sterilizing device may use an ultraviolet LED.

(349) For example, the sterilization unit **333** may be disposed below the first refrigerator compartment cold air auxiliary outlet hole **330**_a_ and the second refrigerator compartment cold air auxiliary outlet hole **330**_b_ in a bar shape that is long in the left-right direction.
(350) The sterilization unit **333** may be disposed on the rear surface of the refrigerator compartment filter and configured to sterilize and discharge the cold air having passed through the refrigerator compartment filter **331** to the refrigerator compartment **51**.
(351) [Connection Duct and Second Storage Chamber Cold Air Supply Duct]
(352) Hereinafter, referring to FIG. **22**, the connection duct **200** and the second storage chamber cold air supply duct **400** will be described in detail.
(353) The connection duct **200** and the second storage chamber cold air duct **400** may be coupled to each other to be one module.
(354) Referring to FIG. **22**C, the connection duct **200** may be formed by coupling a connection duct front plate **210** and a connection duct rear plate **220** to each other.
(355) An upper surface of the connection duct front plate **210** may have an inclined surface that is inclined downward, and a connection duct securing member **213** and a hook securing portion **211** may be formed on the upper surface.
(356) The connection duct securing member **213** and the hook securing portion **211** may be formed in shapes corresponding to the coupling member through hole **313** and the hook **314** of the refrigerator compartment cold air supply duct **300**, respectively, to facilitate smooth coupling between the connection duct **200** and the refrigerator compartment cold air supply duct **300**.
(357) For example, the connection duct front plate **210** and the connection duct rear plate **220** may be easily coupled to each other without a separate coupling member by the hook coupling structure formed along the lateral surface. However, the coupling structure is not limited thereto.
(358) The connection duct rear extended portion **221** may be extended upward from upper end of the connection duct rear plate **220** to be coupled to the refrigerating case **41**.
(359) The connection duct **200** may have one end connected to the grill fan assembly **100** and the other end connected to the refrigerator compartment **51**.
(360) One end of the connection duct **200** may include a refrigerator compartment cold air supply inlet hole **2001** configured to communicate with the grill fan assembly **100**.
(361) The other end of the connection duct **200** may be embedded between the inner case **40** and the outer case **10** on the rear surface of the refrigerator compartment **51**, and configured to communicate with the refrigerator compartment cold air supply duct **300** on the rear surface of the refrigerating case **41**.
(362) Accordingly, the connection duct **200** disposed on the rear surface of the refrigerator compartment **51** may have a minimum thickness in the front-rear direction as possible.
(363) Since one end of the connection duct **200** is connected to the grill fan assembly **100** provided in the freezer compartment **52**, it may be preferred that the connection duct **200** has the minimum thickness in the left-right direction as possible rather than the front-rear direction, in terms of space utilization.
(364) The connection duct **200** may be formed in a shape with the width that becomes narrower from one end to the other end, that is, from the bottom to the top in the front-rear direction.
(365) However, when only the front-rear direction width of the connection duct **200** becomes narrower without changing the left-right direction width of the connection duct **200**, the pressure difference of the cold air flow path might increase.
(366) Accordingly, the connection duct **200** may be formed with the left-right direction width that increases from one end to the other end, that is, from the bottom to the top.
(367) Due to the change in the front-rear direction width and the left-right direction width as described above, the area of the cold air flow path passing through one end of the connection duct **200** and the area of the cold air flow path passing through the other end may as similar as possible.
(368) Since one end of the connection duct has the width that is greater than the other end, one or

more flow path guides **231** may be formed on the connection duct rear plate **220** to prevent turbulence of the cold air flowing into one end and guide the flow of the cold air.
(369) When the connection duct **200** is under pressure by foaming the insulating material, the flow path guide **231** may serve to support the coupling structure of the connection duct front plate **210** and the connection duct rear plate **220**.
(370) The connection duct **200** may include a curved portion that is curved rearward with respect to the front-rear direction so that the other end of the connection duct **200** may be positioned behind one end of the connection duct **200**.
(371) The connection duct **200** includes the curved portion **230** curved in the front-rear direction to pass the outside of the rear surface **41***a* of the refrigerating case, so that it may be embedded in the space between the inner case **40** and the outer case **10** smoothly.
(372) Meanwhile, the second storage chamber cold air supply duct **400** may be formed by coupling a second storage chamber cold air supply duct front plate **410** and a second storage chamber cold air supply duct rear plate **420** to each other.
(373) For example, the second storage chamber cold air supply duct front plate **410** and the second storage chamber cold air supply duct rear plate **420** may be smoothly coupled to each other by a hook coupling structure formed along a lateral surface, even without a separate coupling member, but the coupling type is not limited thereto.
(374) The second storage chamber cold air supply duct **400** may be connected to a support plate **250** formed in a lower area of the connection duct **200** to form one module together with the connection duct **200**.
(375) For example, the support plate **250** may be horizontally extended from one end positioned in the lower area of the connection duct **200**. The support plate **250** may include a second storage chamber duct through hole **251**.
(376) One end of the second storage compartment cold air supply duct **400** may be positioned to communicate with the second storage chamber duct through hole **251** and fixedly fastened to the support plate **250** by using fastening means such as a bolt.
(377) A second storage chamber cold air supply inlet hole **400***a* may be formed in one end of the second storage chamber cold air supply duct **400** disposed to communicate with the second storage chamber duct through hole **251** of the supply plate **250**, and may be configured to communicate with the second storage chamber cold air supply communication hole **45** of the refrigerating case **41** on the rear surface **41***a* of the refrigerating case.
(378) The other end of the connection duct **200** may be connected to the refrigerator compartment **51**.
(379) The other end of the second storage chamber cold air supply duct **400** may be embedded between the inner case **40** and the outer case **10** on the rear surface of the refrigerator compartment **51**, to become in communication with the second storage chamber cold air supply communication hole **45** of the refrigerating case **41** on the rear surface **41***a* of the refrigerating case.
(380) The other end of the second storage chamber cold air supply duct **400** may be formed to direct the second storage chamber cold air supply outlet hole **400***b* supplying cold air to the second storage chamber **51***b* toward the front surface, so that the cold air may be discharged toward the front surface of the refrigerator compartment **51** from the rear surface **41***a* of the refrigerating case.
(381) In addition, a pair of second storage chamber cover coupling members **407** for the coupling with a second storage chamber cold air outlet cover **440** and a temperature sensor inserting portion **409** for inserting a temperature sensor therein may be formed in the other end of the second storage chamber cold air supply duct **400**.
(382) The other end of the second storage chamber cold air supply duct **400** may be formed in a flat plate shape having a wide area toward the front surface so that the member having the various functions described above thereon may be disposed on the flat plate.
(383) Accordingly, the second storage chamber cold air supply duct **300** may be formed in a shape

having a tubular shape from one end toward the other end, that is the bottom to the top, and then having a flat shape at the other end.
(384) The second storage chamber cold air supply duct **400** may be additionally coupled to the lateral surface of the connection duct **200** by using fastening means such as a bolt at the other end having the flat plate shape, to be more strongly coupled.
(385) The second storage chamber cold air supply duct **400** disposed on the rear surface of the refrigerator compartment **51** may have a minimum thickness in the front-rear direction as possible.
(386) However, since one end of the second storage chamber cold air supply duct **400** is connected to the grill fan assembly **100** provided in the freezer compartment **52**, it may be preferred that the second storage chamber cold air supply duct **400** has the minimum thickness in the left-right direction as possible rather than the front-rear direction in terms of space utilization.
(387) Accordingly, the second storage chamber cold air supply duct **400** may be formed in a shape having the front-rear direction width that becomes narrower from one end to the other end, that is, from the bottom to the top.
(388) However, when only the front-rear direction width of the second storage chamber cold air supply duct **400** becomes narrower without changing the left-right direction width, the pressure difference of the cold air flow path might increase.
(389) Accordingly, the second storage chamber cold air supply duct **400** may be formed with the left-right direction width that increases from one end to the other end, that is, from the bottom to the top.
(390) Due to the change in the front-rear direction width and the left-right direction width as described above, the area of the cold air flow path passing through one end of the second storage chamber cold air supply duct **400** and the area of the cold air flow path passing through the other end may as similar as possible.
(391) The second storage chamber cold air supply duct **400** may include a curved portion **430** that is curved rearward with respect to the front-rear direction so that the other end of the connection duct **200** may be positioned behind one end of the connection duct **200**.
(392) The second storage chamber cold air supply duct **400** includes the curved portion **430** curved in the front-rear direction to pass the outside of the rear surface **41***a* of the refrigerating case, so that it may be embedded in the space between the inner case **40** and the outer case **10** smoothly.
(393) [Refrigerator Compartment Cold Air Returning Duct]
(394) Hereinafter, referring to FIG. **23**, the refrigerator compartment cold air returning duct **500** will be described in detail.
(395) The refrigerator compartment cold air returning duct **500** may be formed by coupling a refrigerator compartment cold air returning duct front plate **510** and refrigerator compartment cold air returning duct rear plate **520** to each other.
(396) For example, the refrigerator compartment cold air returning duct front plate **510** and the refrigerator compartment cold air returning duct rear plate **520** may be easily coupled to each other without a separate coupling member by a hook coupling structure formed along the lateral surface. However, the coupling structure is not limited thereto.
(397) The refrigerator compartment cold air returning duct **500** may have one end connected to the rear surface **41***a* of the freezing case and the other end connected to the lower surface **41***c* of the refrigerating case.
(398) Specifically, a refrigerator compartment cold air returning outlet hole **500***b* may be formed in one end of the refrigerator compartment cold air returning duct toward the front surface, to facilitate communication between the end and the rear surface **42***a* of the freezing case.
(399) The refrigerator compartment cold air returning outlet hole **500***b* of the refrigerator compartment cold air returning duct **500** may be in communication with the refrigerator compartment cold air returning communication outlet hole **46***b* formed on the rear surface **42***a* of the freezing case.

(400) A refrigerator compartment cold air returning inlet hole **500***a* may be formed upward in the other end of the refrigerator compartment cold air returning duct **500**, to facilitate communication between the other end and the rear surface **41***a* of the refrigerating case.
(401) The refrigerator compartment cold air returning inlet hole **500***a* of the refrigerator compartment cold air returning duct **500** may be in communication with the refrigerator compartment cold air returning communication inlet hole **46***a* formed on the lower surface **41***c* of the refrigerating case.
(402) A returning duct rear extended portion **521** may be vertically extended from the other end of the refrigerator compartment cold air returning duct **500** along an upper rear end of the refrigerator compartment cold air returning duct **500**.
(403) In addition, a returning duct side extended portion **522** may be vertically extended from the other end of the refrigerator compartment cold air returning duct **500** along an upper end of the refrigerator compartment cold air returning duct **500**.
(404) The returning duct rear extended portion **521** and the returning duct side extended portion **522** may be connected to each other.
(405) Accordingly, the other end of the refrigerator compartment cold air returning duct **500** may be fixed to the refrigerating case **41** to support the lower surface **41***c* and the rear surface **41***a* of the refrigerating case.
(406) A returning duct filter **530** may be disposed in the refrigerator compartment cold air returning duct **500** and configured to allow the cold air returning by the refrigerator compartment cold air returning duct **500** to pass therethrough.
(407) The returning duct filter **530** may be a deodorizing filter including activated carbon to remove odors.
(408) In the present disclosure, no separate suction fan for sucking and deodorizing cold air in a target region is provided but the overall cold air flow path system circulating inside the refrigerator **1** may deodorize the cold air. Due to this structure, the smell of the refrigerator compartment **51** and the smell of the freezer compartment **52** could be mixed with each other.
(409) Accordingly, the cold air flowing into the refrigerator compartment **51** may be primarily deodorized by the refrigerator compartment filter **331** provided in the refrigerator compartment cold air supply duct **300**.
(410) After that, the cold air re-flowing into the freezer compartment **52** after passing through the refrigerator compartment **51** may be secondarily deodorized by the returning duct filter **530** provided in the refrigerator compartment cold air returning duct **500**, to reduce the mixture of the smells in the refrigerator compartment **51** and the freezer compartment **52**.
(411) With respect to the left-right direction of the refrigerator compartment cold air returning duct **500**, the connection duct **200** may be disposed in an area next to the refrigerator compartment cold air returning duct **500** and the ice-making chamber cold air supply duct **600** may be disposed in other area.
(412) Since the connection duct **200** may be disposed in the area next to the refrigerator compartment cold air returning duct **500** and the ice-making chamber cold air supply duct **600** is disposed on the other area with respect to the refrigerator compartment cold air returning duct **500**, the entire cold air balance of the refrigerator **1** based on the refrigerator compartment cold air supply duct **500** may be maintained.
(413) [Grill Fan Assembly]
(414) Hereinafter, referring to FIGS. **24** to **27**, the grill fan assembly **100** will be described in detail.
(415) The grill fan assembly **100** according to the present disclosure may include a shroud **120** and a grill fan **110**.
(416) The shroud **120** may define a rear exterior design of the grill fan assembly **100** and the grill fan **110** may define a front exterior design of the grill assembly **100**.
(417) The grill fan **110** may be disposed toward the front surface of the freezer compartment **52**,

and the shroud **120** may be disposed toward the rear surface **42***a* of the freezing case, that is, the evaporator **101** provided on the rear wall of the freezing case.
(418) The shroud **120** may include a first inlet hole **121***a* and a second inlet hole **121***b*.
(419) The cold air heat-exchanged while passing through the evaporator **101** disposed behind the shroud **120** may flow into the space formed between the first inlet hole **121***a* and the second inlet hole **121***b*.
(420) A freezing fan module **160** is disposed on a front surface of the first inlet hole **121***a* and an ice-making fan **170** may be disposed on a front surface of the second inlet hole **121***b*.
(421) The first inlet hole **121***a* may be provided in an upper center region of the grill fan assembly **100** and the second inlet hole **121***b* may be provided in one side region of the grill fan assembly **100** with respect to the first inlet hole **121***a*.
(422) Since the ice-making fan module **170** is disposed the second inlet hole **121***b* to supply cold air to the ice-making chamber **22**, the second inlet hole **121***b* may be disposed adjacent to the other lateral surface **42***e* of the freezing case where the ice-making chamber cold air supply duct **600** is provided.
(423) With respect to the first inlet hole **121***a*, a flow path opening/closing module seating portion **122** on which the flow path opening/closing module **130** is seated may be formed in the other side region rather than the side region of the grill fan assembly **100** in which the second inlet hole **121***b* is disposed.
(424) The flow path opening/closing module **130** may include a flow path opening/closing damper **140** and **150** configured to selectively cut off the cold air supplied to the refrigerator compartment **51**.
(425) The refrigerator compartment **51** may include a first storage chamber **51***a* and a second storage chamber **51***b* that are preset to have different temperatures, respectively.
(426) In this instance, the flow path opening/closing module **130** may include a first storage chamber flow path opening/closing damper **140** for selectively cutting off the cold air supplied to the first storage chamber **51***a* and a second storage chamber flow path opening/closing damper **150** for selectively cutting off the cold air supplied to the second storage chamber **51***b*.
(427) The first storage chamber flow path opening/closing damper **140** and the second storage chamber flow path opening/closing damper **150** may be seated on the flow path opening/closing module seating portion **122**, in a state of being covered by a damper cover **131**.
(428) The damper cover **131** may be formed of an insulating material such as Styrofoam, and the material is not limited thereto.
(429) The damper cover **131** may be formed by coupling a first damper cover **131***a*, a second damper cover **131***b* and a third damper cover **131***c* to each other.
(430) The first storage chamber flow path opening/closing damper **140** may be disposed between the first damper cover **131***a* and the second damper cover **131***b*, and the second storage chamber flow path opening/closing damper **150** may be disposed between the second damper **131***b* and the third damper **131***c*.
(431) A first storage chamber cold air outlet **132***a* may be formed on an upper surface of the damper cover **131** covering the first storage chamber flow path opening/closing damper **140** to be in communication with the connection duct **200** to supply cold air.
(432) A second storage chamber cold air outlet **132***b* may be formed on an upper surface of the damper cover **131** covering the second storage chamber flow path opening/closing damper **150** to be in communication with the second storage chamber cold air supply **400** to supply cold air.
(433) With respect to the left-right direction of the grill fan assembly **100**, the freezing fan module **160** may have an area overlapping with the first storage chamber flow path opening/closing damper **140** that is larger than an area overlapping with the second storage chamber flow path opening/closing damper **150**.
(434) For example, an area of the flow path opening/closing module seating portion **122**

corresponding to the second storage chamber flow path opening/closing damper **150** may be more projected toward a rear surface of the shroud than an area thereof corresponding to the first storage chamber flow path opening/closing damper **140**.
(435) The first storage chamber **51***a* and the second storage chamber **51***b* may be adjusted to have different amounts of cold air to be adjusted at different temperatures, respectively.
(436) The cold air guided to the flow path opening/closing module **130** may be branched from the first storage chamber flow path opening/closing damper **140** and the second storage chamber flow path opening/closing damper **150** to be supplied to the first storage chamber **51***a* and the second storage chamber **51***b*, respectively.
(437) In this instance, since the first storage chamber flow path opening/closing damper **140** overlaps more in the left-right direction with the freezing fan module **160** configured to blow cold air, the cold air blown by the freezing fan module **160** may be guided more toward the first storage chamber flow path opening/closing damper **140**.
(438) As described above, since the area of the freezing fan module **160** overlapping with the first storage chamber flow path opening/closing damper **140** is larger than the area thereof overlapping with the second storage chamber flow path opening/closing damper **150**, it may be possible to induce a cold air flow capable of supplying more cold air to the second storage chamber **51***b* than the first storage chamber **51***a*.
(439) A water discharge hole **129** configured to discharge a condensate generated by a temperature difference between the spaces may be further formed in a lower center region of the shroud **120**.
(440) The grill fan **110** disposed on the front surface of the shroud **120** may be coupled to the shroud **120**, to accommodate the ice-making fan module **170**, the freezing fan module **160** and the flow path opening/closing module **130**.
(441) A grill fan upper region outlet hole **111** may be formed in an upper center region of the grill fan **110** and configured to discharge the cold air blown by the freezing fan module **160** toward an upper front surface of the freezer compartment **52**.
(442) In this instance, some of the cold air blown by the ice-making fan module **170** may be discharged to the freezer compartment **52** through the grill fan upper region outlet hole **111**.
(443) A grill fan lower region outlet hole **112***a* may be formed in a lower center region of the grill fan **110** to discharge the cold air blown by the freezing fan module **160** toward a lower front surface of the freezer compartment **52**.
(444) The grill fan lower region outlet hole **112** may be provided with a first grill fan lower region outlet hole **112***a* and a pair of second grill fan lower region outlet holes **112***b* disposed on both lateral surfaces with respect to the first grill fan lower region outlet hole **112***a*.
(445) The second grill fan lower region outlet hole **112***b* may guide the cold air discharged to the freezer compartment **52** to flow to the both lateral surfaces to uniformly circulate the overall region of the freezer compartment **52**.
(446) A pair of freezer compartment cold air returning guides **119** may be formed below the grill fan lower region outlet holes **112***a* and **112***b* to guide the returning cold air.
(447) The cold air having circulated the ice-making chamber **22** and the cold air having circulated the freezer compartment **52** may return to the freezer compartment cold air returning guide **119** provided in the lower region of the freezer compartment **52** to be supplied to the evaporator **101**.
(448) The cold air blown by the freezing fan module **160** may be supplied to the ice-making chamber **22**, the refrigerator compartment **51** including the first storage chamber **51***a* and the second storage chamber **51***b*, and the freezer compartment **52**.
(449) The cold air blown by the ice-making fan module **170** may be supplied to the ice-making chamber **22** and the freezer compartment **52**.
(450) Accordingly, the refrigerator **1** according to the present disclosure may supply cold air all of the refrigerator compartment **51**, the freezer compartment **52** and the ice-making chamber **22** by using one evaporator **101** and one grill fan assembly **100**.

(451) The refrigerator **1** may adjust the cold air supplied to the refrigerator compartment **51** by using the flow path opening/closing module **130** provided in the grill fan assembly **100**, thereby providing a new cold air supply system.
(452) If the flow path opening/closing module **130** is disposed in the refrigerator compartment **51**, not the freezer compartment **52**, the inner volume of the refrigerator compartment **51** may be reduced as much as the space occupied by the flow path opening/closing module **130**.
(453) According to the present disclosure, since the flow path opening/closing module **130** for selectively cutting off the cold air supplied to the refrigerator compartment **51** is provided in the grill fan assembly **100** blowing the cold air generated by one evaporator **101** to the refrigerator compartment **51** and the freezer compartment **52**, there may be no need of securing an additional separate space for the flow path opening/closing module **130** in the refrigerator compartment **51**.
(454) Accordingly, the refrigerator **1** may provide a new cold air supply system capable of enhancing capacity competitiveness of the refrigerator **1**.
(455) The refrigerator **1** having the new cold air supply system capable of enhancing the capacity competitiveness may include one grill fan assembly **100** disposed in the freezer compartment **52** and configured to supply cold air to all of the ice-making chamber **22**, the refrigerator compartment **51** and the freezer compartment **52**, and two flow path opening/closing dampers **140** and **150** configured to selectively cut off the cold air supplied to the refrigerator compartment **51** as one embodiment. However, embodiments of the present disclosure are not limited thereto.
(456) For example, referring to FIGS. **25**, **26**A and **26**B, one grill fan assembly **100** may be configured to supply cold air to the ice-making chamber **22**, the refrigerator compartment **51** and the freezer compartment **52**, and may include one flow path opening/closing damper **140** configured to selectively cut off the cold air supplied to the refrigerator compartment **51**.
(457) Specifically, the grill fan assembly **100** according to another embodiment may include the ice-making fan module **170**, the freezing fan module **160** and one refrigerator compartment flow path opening/closing damper **140** that are provided between the shroud and the grill fan **110**.
(458) The freezing fan module **160** may be disposed in the first inlet hole **121***a*, and the ice-making fan module **170** may be disposed in the second inlet hole **121***b* of the shroud **120**.
(459) The refrigerator compartment flow path opening/closing damper **140** may be covered by the first damper cover **131***a* and the second damper cover **131***b*.
(460) An ice-making chamber cold air outlet guide **173** including an ice-making chamber outlet hole **172** may be disposed on the ice-making fan module **170** and configured to guide the cold air supplied in communication with the ice-making chamber cold air supply duct **600**.
(461) A refrigerator compartment cold air outlet guide **133** may be disposed on the refrigerator compartment flow path opening/closing module **130** including the refrigerator compartment flow path opening/closing damper **140**, the first damper cover **131***a* and the second damper cover **131***b*, and may be configured to guide the cold air supplied in communication with the connection duct **200**.
(462) For example, referring to FIGS. **25**, **27**A and **27**B, one grill fan assembly **100** may be configured to supply cold air to the refrigerator compartment **51** and the freezer compartment **52**, and may include one flow path opening/closing damper **140** configured to selectively cut off the cold air supplied to the refrigerator compartment **51**.
(463) Specifically, the grill fan assembly **100** according to this embodiment may include a flow path cut-off member **180** disposed between the shroud **120** and the grill fan **110**, and one refrigerator compartment flow path opening/closing damper **140**.
(464) The freezing fan module **160** may be disposed in the first inlet hole **121***a* of the shroud **120** and the flow path cut-off member **180** may be disposed in the second inlet hole **121***b* of the shroud **120**.
(465) The flow path cut-off member **180** may have a shape capable of cutting off the cold air supplied to the second inlet hole **121***b*.

(466) Since the flow path cut-off member **180** is disposed at a position where the ice-making fan module **170** is provided, the cold air may be prevented from leaking through the ice-making chamber cold air outlet guide **173**.
(467) Accordingly, in case of a refrigerator model without the ice-making chamber **22**, the grill fan assembly **100** may be used without a separate shape change only by simply installing the flow path cut-off member **180**, thereby enhancing assembly process efficiency.
(468) The refrigerator compartment flow path opening/closing damper **140** may be covered by the first damper cover **131***a* and the second damper cover **131***b*.
(469) A refrigerator compartment cold air outlet hole guide **133** including a refrigerator compartment cold air outlet hole **132** may be disposed on the refrigerator compartment flow path opening/closing module **130** including the refrigerator compartment flow path opening/closing damper **140**, the first damper cover **131***a* and the second damper cover **131***b*, and may be configured to guide the cold air in communication with the connection duct **200**.
(470) According to the present disclosure, since the flow path opening/closing damper **140** configured to selectively cut off the cold air supplied to the refrigerator compartment **51** is provided in the grill fan assembly **100** configured to blow the cold air generated by one evaporator **101** to the refrigerator compartment **51** and the freezer compartment **52**, there may be no need of securing an additional separate space for the flow path opening/closing module **130** in the refrigerator compartment **51**.
(471) Accordingly, the refrigerator **1** may provide a new cold air supply system capable of enhancing capacity competitiveness of the refrigerator **1**.
(472) The embodiments are described above with reference to a number of illustrative embodiments thereof. However, the present disclosure is not intended to limit the embodiments and drawings set forth herein, and numerous other modifications and embodiments can be devised by one skilled in the art. Further, the effects and predictable effects based on the configurations in the disclosure are to be included within the range of the disclosure though not explicitly described in the description of the embodiments.

Claims

1. A refrigerator comprising: an inner case comprising (i) a refrigerating case that defines a refrigerator compartment and (ii) a freezing case that defines a freezer compartment; an outer case disposed outside the inner case; a connection duct disposed between the inner case and the outer case; and a refrigerator compartment cold air supply duct that is detachably coupled to a rear inner surface of the refrigerating case and in fluid communication with the connection duct, wherein the connection duct and the refrigerator compartment cold air supply duct define a cold air flow path extending in an up-down direction, wherein an upper surface of the connection duct is coupled to and faces a lower surface of the refrigerator compartment cold air supply duct, wherein the upper surface of the connection duct and the lower surface of the refrigerator compartment cold air supply duct are inclined with respect to a surface of the refrigerator compartment, wherein the refrigerating case defines a duct inserting groove at an upper surface of the refrigerating case, and wherein an upper area of the refrigerator compartment cold air supply duct is inserted in and coupled to the duct inserting groove.
2. The refrigerator of claim 1, further comprising an insulating material that is disposed between the inner case and the outer case and in direct contact with the connection duct.
3. The refrigerator of claim 1, wherein the cold air flow path has a linear shape.
4. The refrigerator of claim 1, wherein a portion of a front surface of the refrigerator compartment cold air supply duct is parallel to a portion of a front surface of the connection duct.
5. The refrigerator of claim 1, wherein the inner case has a refrigerator compartment cold air supply communication hole that is defined at a rear surface of the inner case and open upward.

6. The refrigerator of claim 5, wherein the refrigerating case comprises a rear projected portion that protrudes from a rear surface of the refrigerating case toward an inside of the refrigerating case, and wherein the refrigerator compartment cold air supply communication hole is defined at an upper surface of the rear projected portion.
7. The refrigerator of claim 6, wherein at least a portion of the connection duct is inserted in the rear projected portion from an outer surface of the refrigerating case.
8. The refrigerator of claim 6, wherein the connection duct is coupled to the refrigerating case and supports a rear surface of the rear projected portion.
9. The refrigerator of claim 1, wherein the connection duct comprises a rear extended portion that extends vertically from a rear end of the upper surface of the connection duct, the rear extended portion being coupled to the refrigerating case and supporting a rear surface of the refrigerating case.
10. The refrigerator of claim 1, wherein the upper surface of the connection duct and the lower surface of the refrigerator compartment cold air supply duct are inclined in a direction descending toward a front surface of the refrigerator compartment, and wherein the upper surface of the connection duct and the lower surface of the refrigerator compartment cold air supply duct are coupled to each other by sliding on each other.
11. The refrigerator of claim 1, wherein the duct inserting groove extends along a boundary between the upper surface of the refrigerating case and a rear surface of the refrigerating case.
12. The refrigerator of claim 1, wherein the refrigerator compartment cold air supply duct comprises: a duct insulating portion that defines the cold air flow path; and a duct sheet coupled to a rear side of the duct insulating portion and covers the cold air flow path, and wherein a thickness of the duct sheet is less than a thickness of the duct insulating portion.
13. The refrigerator of claim 1, wherein the connection duct has a first end connected to the freezer compartment and a second end connected to the refrigerator compartment, the connection duct passing through a center of the refrigerator compartment with respect to a left-right direction, and wherein a width of the cold air flow path increases from the first end to the second end of the connection duct.
14. The refrigerator of claim 1, further comprising: a door configured to open and close at least a portion of the refrigerator compartment; an ice-making chamber defined in the door; and an ice-making chamber cold air supply duct configured to supply cold air to the ice-making chamber, wherein the ice-making chamber cold air supply duct passes through a lateral surface of the refrigerator compartment.
15. The refrigerator of claim 14, wherein the refrigerator compartment cold air supply duct defines a first refrigerator compartment cold air flow path and a second refrigerator compartment cold air flow path that are configured to branch cold air from the connection duct, and wherein the first refrigerator compartment cold air flow path is wider than the second refrigerator compartment cold air flow path, and wherein the ice-making chamber cold air supply duct is disposed closer to the second refrigerator compartment cold air flow path than to the first refrigerator compartment cold air flow path.
16. The refrigerator of claim 14, further comprising: a refrigerator compartment cold air returning duct configured to guide cold air discharged from the refrigerator compartment, the refrigerator compartment cold air returning duct having: a first end that is connected to the refrigerator compartment and passes through a center of the refrigerator compartment with respect to a left-right direction, and a second end that is connected to the freezer compartment and passes through a center of the freezer compartment with respect to the left-right direction.
17. The refrigerator of claim 16, wherein the refrigerator compartment comprises a first storage chamber and a second storage chamber, and wherein the refrigerator further comprises a storage chamber cold air supply duct disposed at an outer surface of the refrigerating case and configured to supply cold air to the second storage chamber, wherein the connection duct and the storage

chamber cold air supply duct are disposed at a first side of the refrigerator compartment cold air returning duct with respect to the left-right direction, and wherein the ice-making chamber cold air supply duct is disposed at a second side of the refrigerator compartment cold air returning duct with respect to the left-right direction.

18. The refrigerator of claim 17, further comprising an insulating material that is disposed between the inner case and the outer case and in direct contact with the storage chamber cold air supply duct.

19. The refrigerator of claim 17, further comprising: a first storage chamber flow path damper that is disposed in the freezer compartment and configured to selectively block cold air from being supplied to the connection duct; and a second storage chamber flow path damper that is disposed in the freezer compartment and configured to selectively block cold air from being supplied to the storage chamber cold air supply duct.

20. The refrigerator of claim 1, wherein the refrigerator compartment cold air supply duct defines: a refrigerator compartment cold air main outlet hole configured to discharge cold air to the refrigerator compartment; and a refrigerator compartment cold air auxiliary outlet hole, and wherein the refrigerator compartment cold air supply duct comprises: a refrigerator compartment cold air auxiliary outlet guide disposed at a front side of the refrigerator compartment cold air auxiliary outlet hole, and a refrigerator compartment filter disposed at the refrigerator compartment cold air auxiliary outlet guide and configured to filter cold air supplied through the refrigerator compartment cold air auxiliary outlet hole.

21. The refrigerator of claim 20, wherein the refrigerator compartment cold air auxiliary outlet guide comprises an insulating member spaced apart from a front surface of the refrigerator compartment filter, and wherein the refrigerator compartment cold air auxiliary outlet guide is configured to guide the cold air having passed through the refrigerator compartment filter to be discharged downward to the refrigerator compartment through a space defined between the insulating member and the refrigerator compartment filter.

22. The refrigerator of claim 16, further comprising a returning duct filter disposed inside the refrigerator compartment cold air returning duct and configured to filter cold air guided by the refrigerator compartment cold air returning duct.

23. The refrigerator of claim 1, wherein an upper end of the upper surface of the connection duct extends upward and is attached to a rear surface of the refrigerator compartment cold air supply duct, wherein a lower end of the upper surface of the connection duct extends downward and is located forward relative to the upper end of the upper surface of the connection duct, and wherein the upper surface of the connection duct and the lower surface of the refrigerator compartment cold air supply duct are inclined with respect to the rear surface of the refrigerator compartment cold air supply duct and extend downward and forward toward a front surface of the refrigerator compartment.

24. The refrigerator of claim 1, wherein the duct inserting groove is disposed adjacent to a ceiling of the refrigerator compartment.